MANCHESTER 1824
The University of Manchester

EART11100

How to Build a Planet

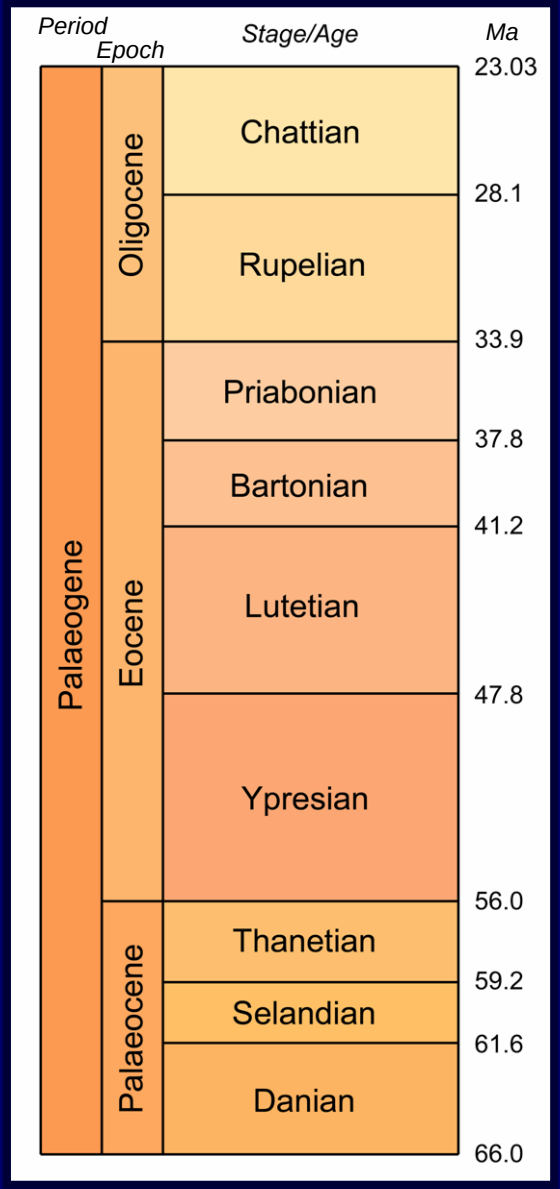
Class 11

The Cenozoic Era: Our Earth

66.0 Ma – 0 Ma

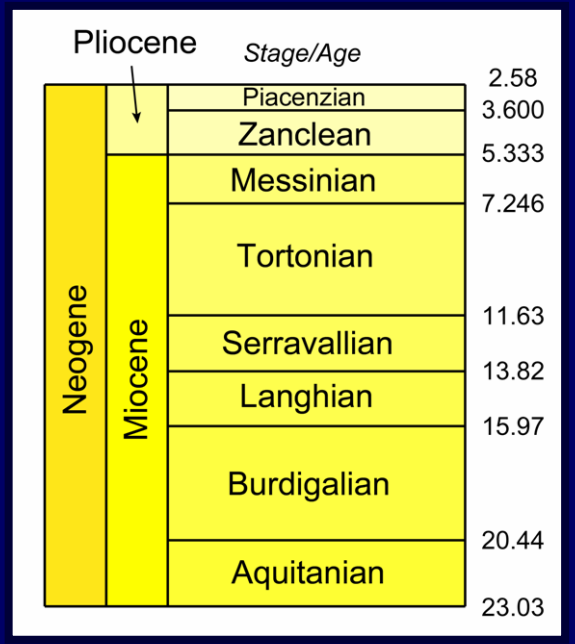
- list in order the periods and epochs of Cenozoic and state their ages
- name the major climatic events of the Cenozoic in the order in which they occurred, and locate them on a temperature or sea level curve
- describe the causes of major climatic events in terms of plate tectonics, changing ocean circulation patterns and atmospheric composition

About the Cenozoic (66 Ma – 0): the age of recent life

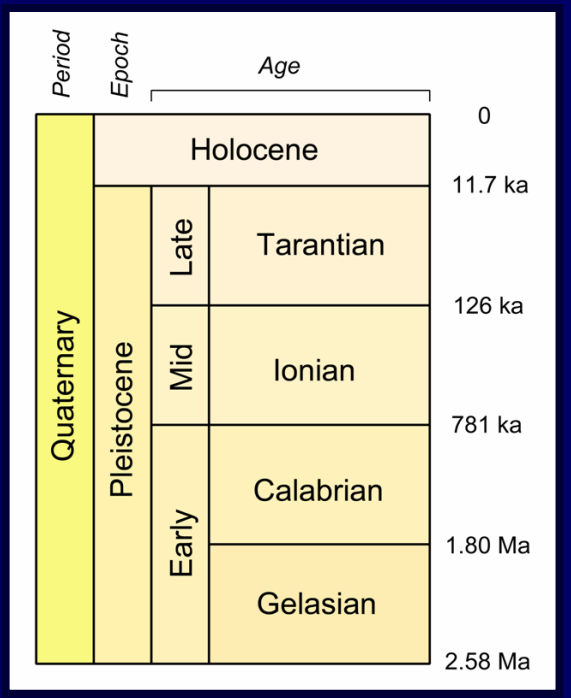


Palaeogene = old birth

- Cenozoic (aka Cainozoic, Caenozoic) era incorporates the Palaeogene, Neogene, and Quaternary periods
- Palaeozoic (aka Primary) and Mesozoic (aka Secondary)
- Tertiary = Palaeogene + Neogene

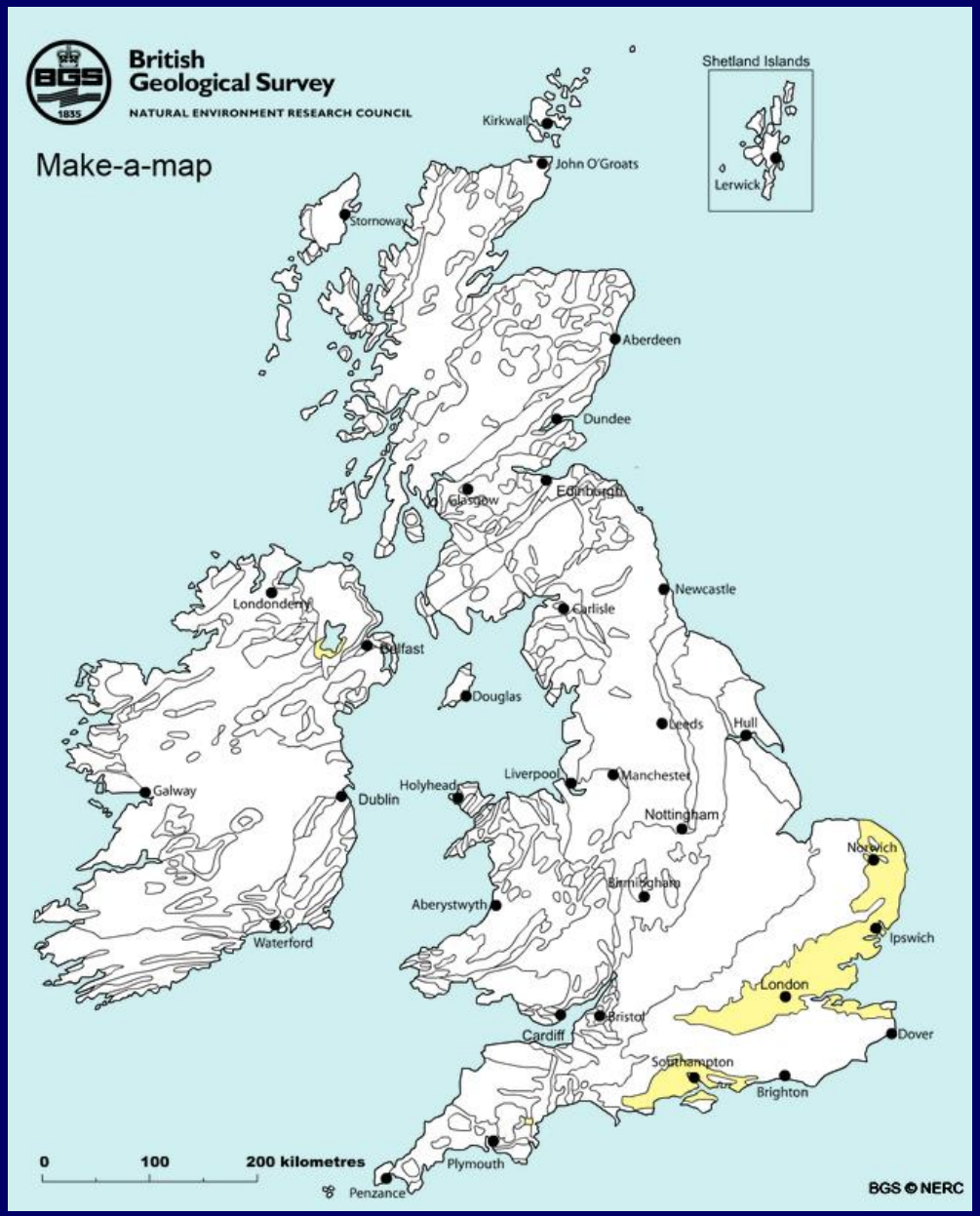


Neogene = new birth

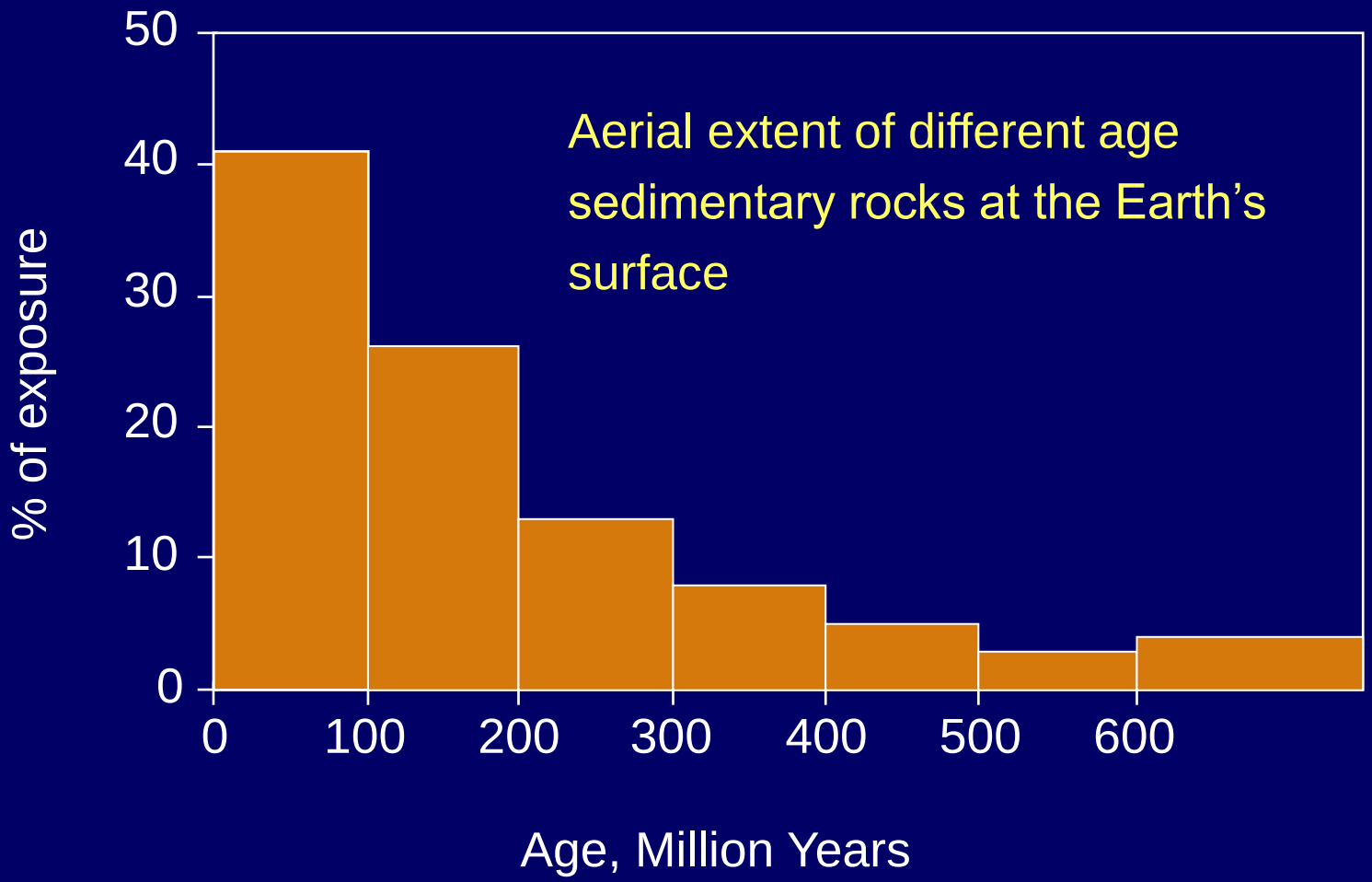


Quaternary = fourth period

Cenozoic rock outcrops in the UK

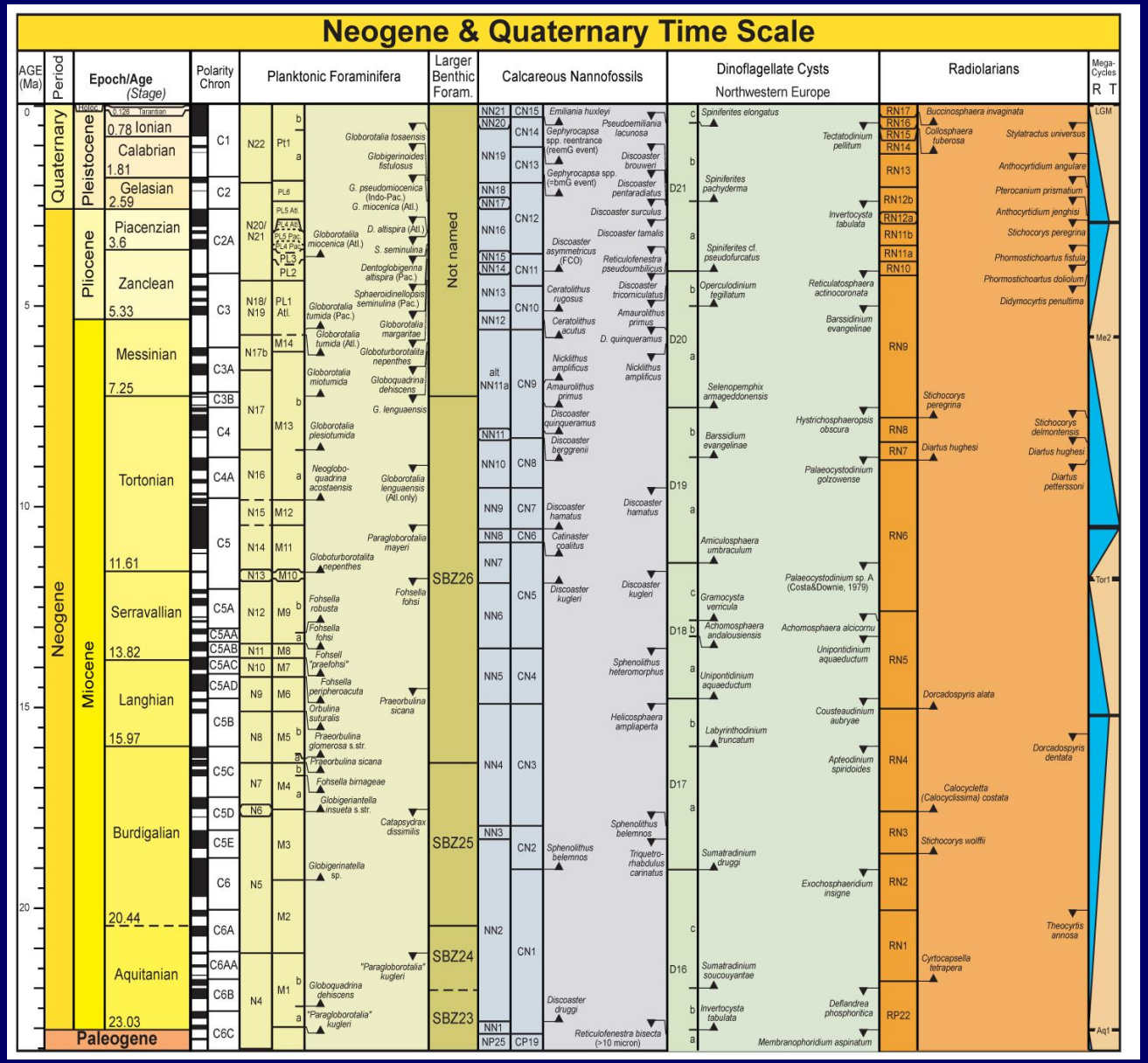


The Cenozoic sedimentary record is relatively complete

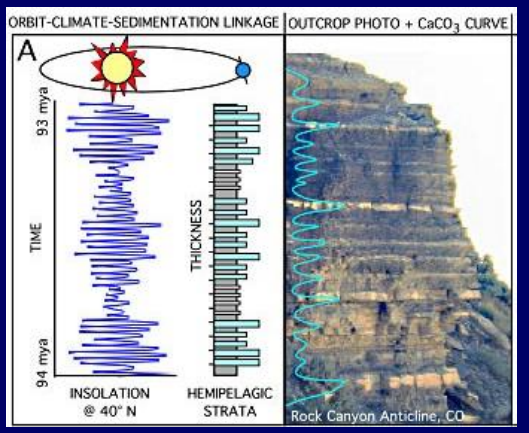


Cenozoic stratigraphic marine and terrestrial record is more complete and dated at a much higher resolution than the earlier eras of the Phanerozoic, particularly as we get closer to the present day

Subdivision of the Cenozoic time scale



Cyclostratigraphy



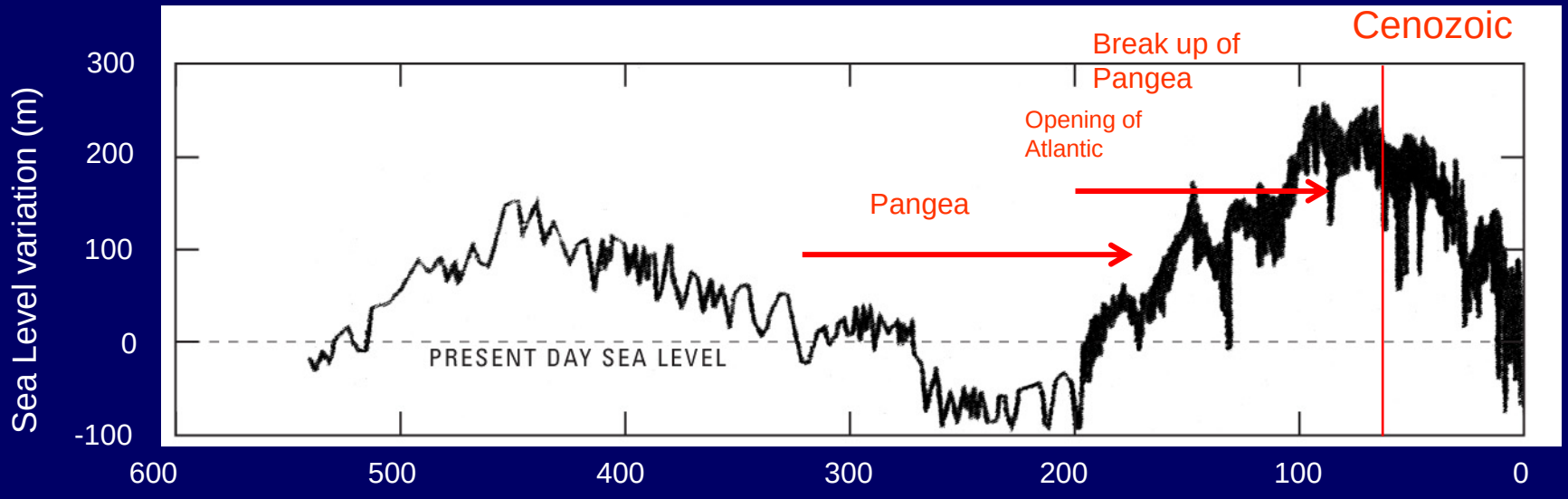
Sedimentary record of orbital influence

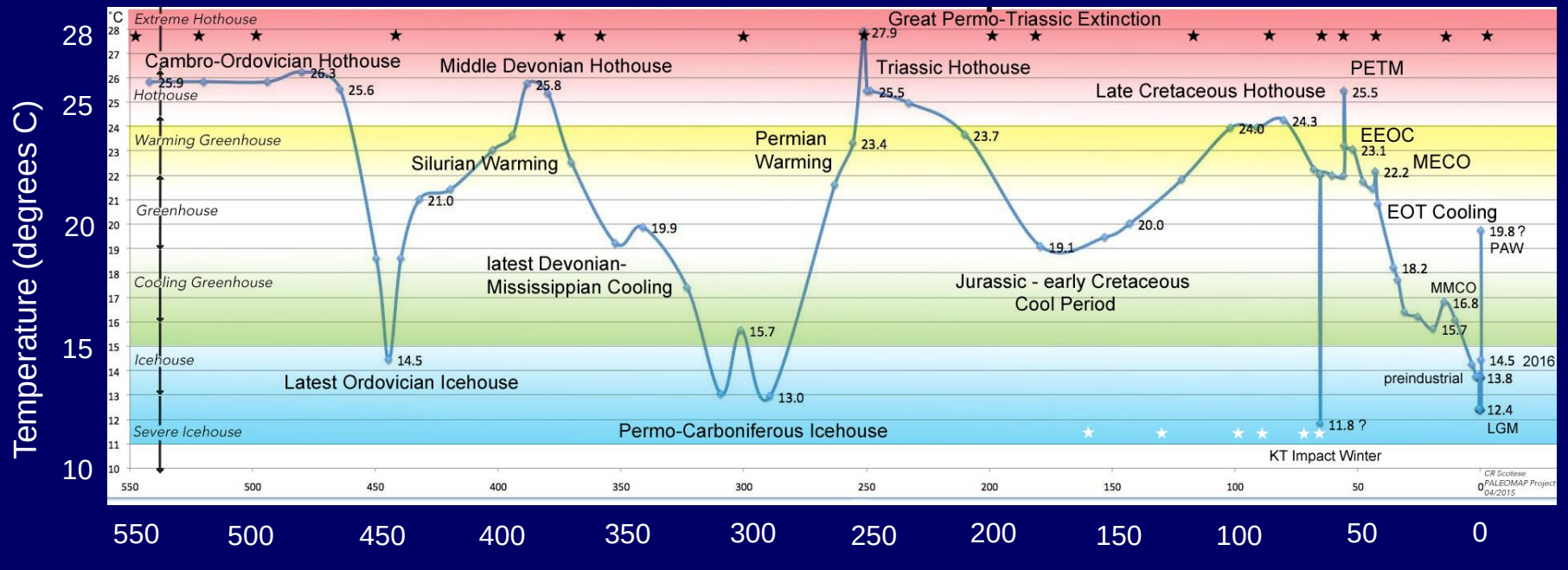
Calcareous nannofossils
Planktonic and benthic foraminifera
Radioalaria
Dinofalgellate cysts (single-cell algae)



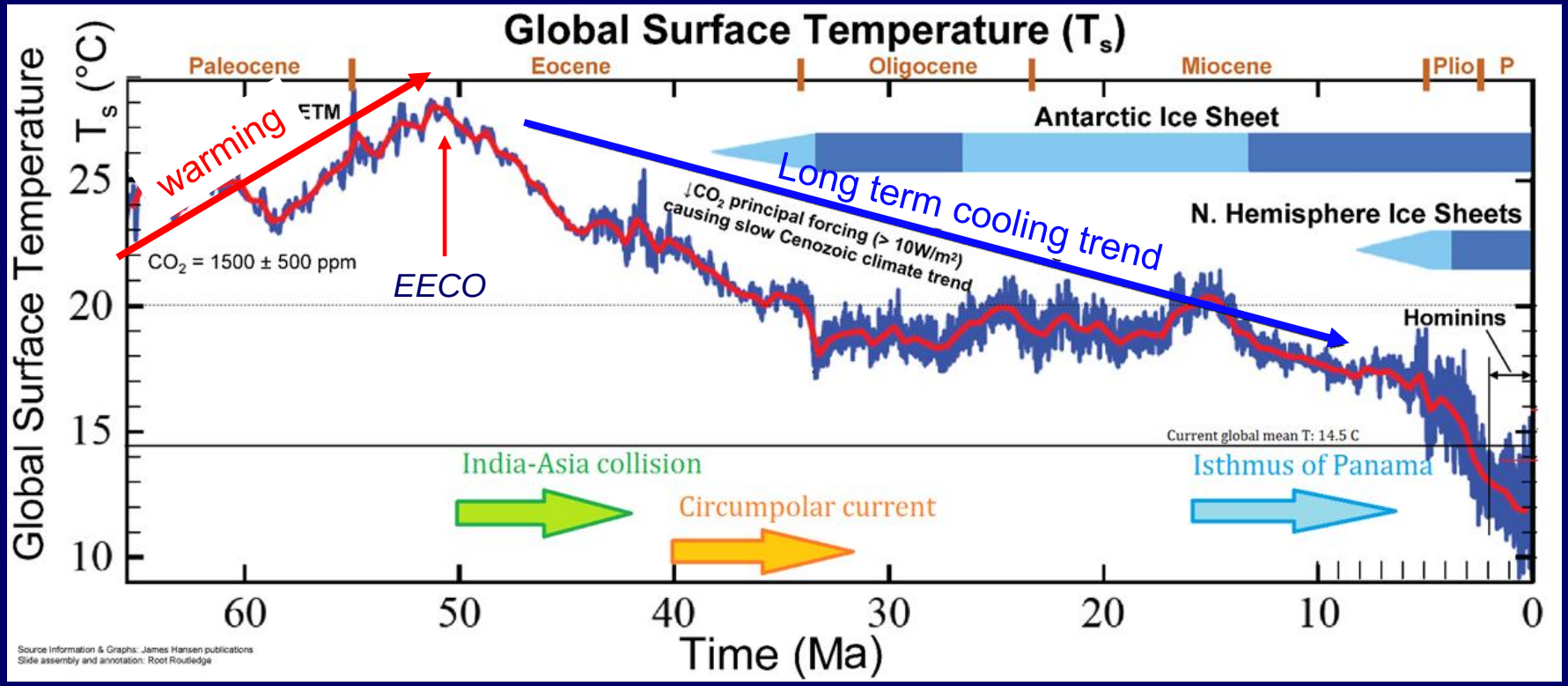
Optical micrograph of various foraminifera

Phanerozoic context of Cenozoic climate and sea level change



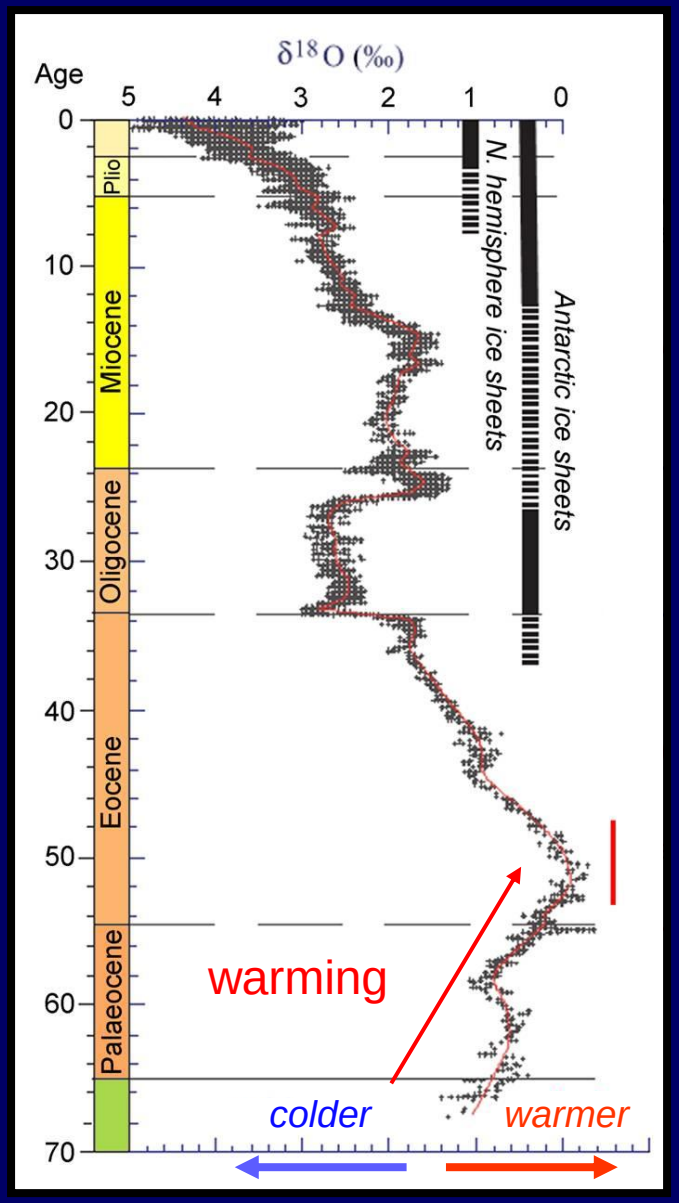


Overview of Cenozoic global temperature (and climate)



- The early Palaeocene - mid-Eocene (~50 Ma) warming trend culminating in the Early Eocene Climatic Optimum (EECO) 50 million years ago,
- A subsequent long-term cooling phase culminating in the formation of the polar ice sheets and ice age that continues to the present day.

Palaeocene warming trend & Early Eocene climatic optimum



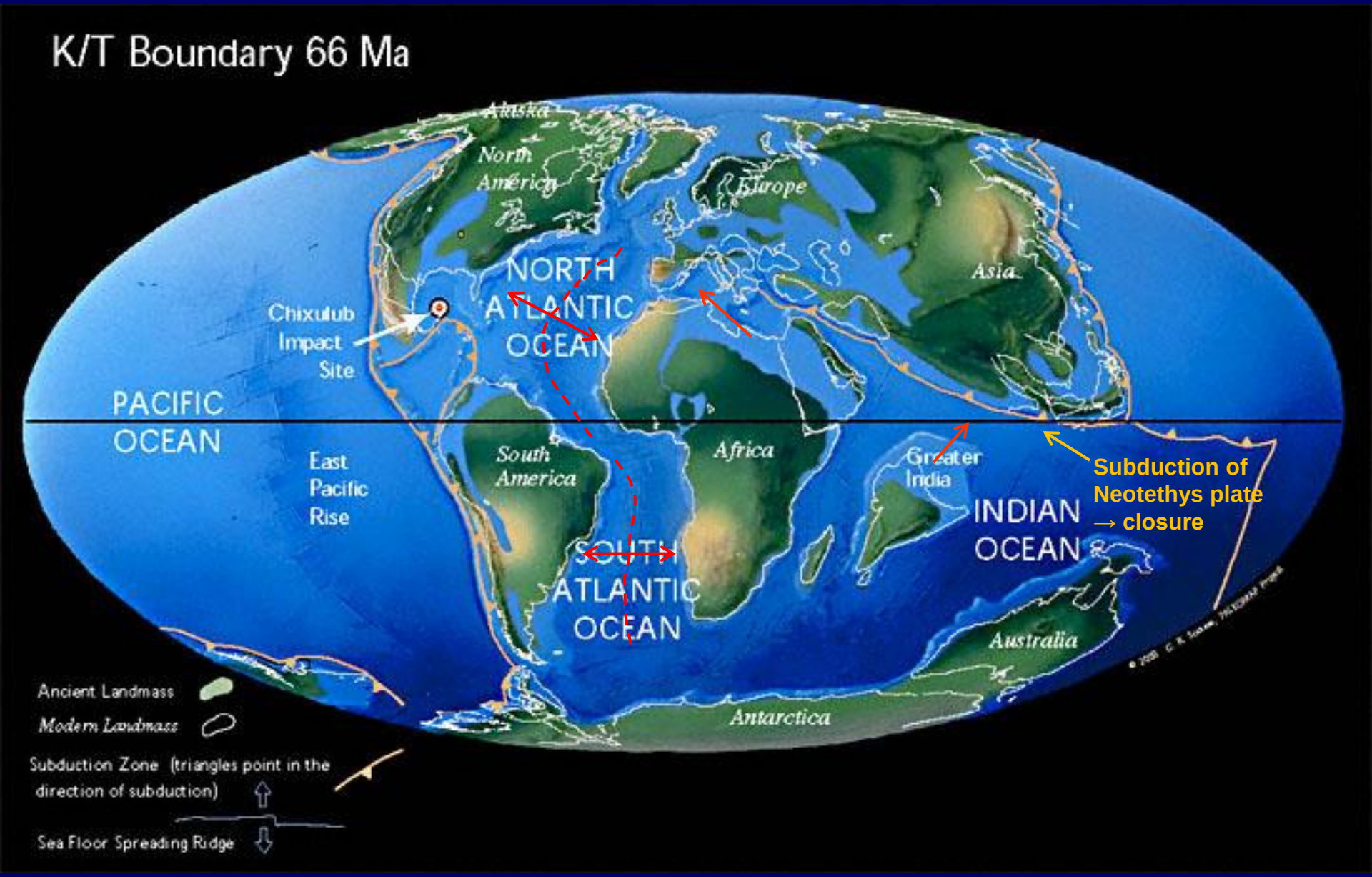
$\delta^{18}O$ data were obtained primarily from analysis of benthic microfossils

	$\delta^{18}O$	
	More Negative	More Positive
Benthic Foraminifera	-Warmer global climate -Decreased ice volume	-Colder global climate -Increased ice volume
Planktic Foraminifera	-Warmer local temperatures -Decreased ice volume -Increased local precipitation	-Colder local temperatures -Increased ice volume -Increased local evaporation

Early Eocene Climatic Optimum
(tropical waters 5-10 ℃ warmer, deep oceans 9-12 ℃ warmer)

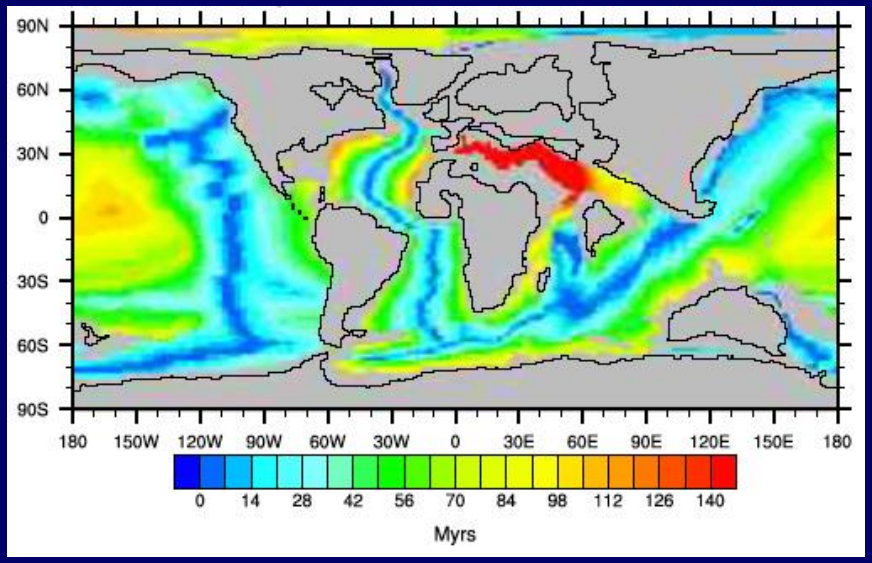
warming trend → changes in configuration of continents

Palaeocene / Eocene palaeogeography

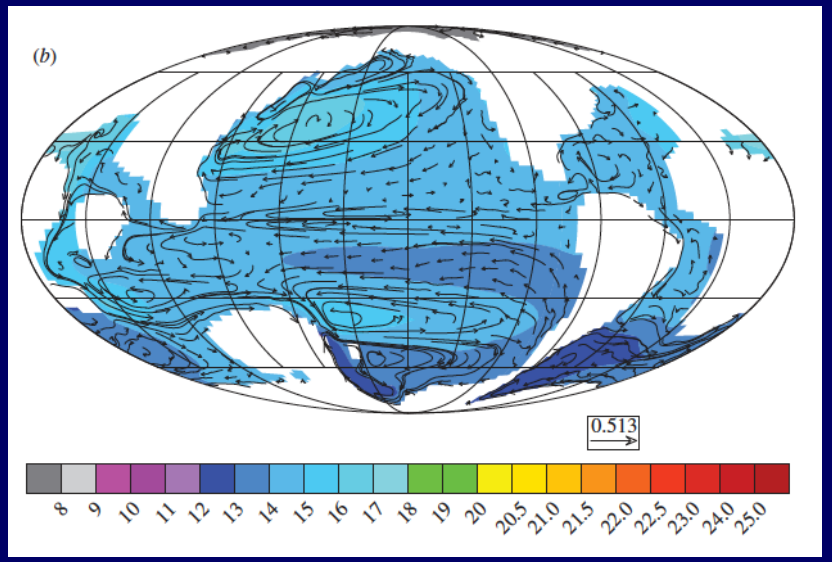


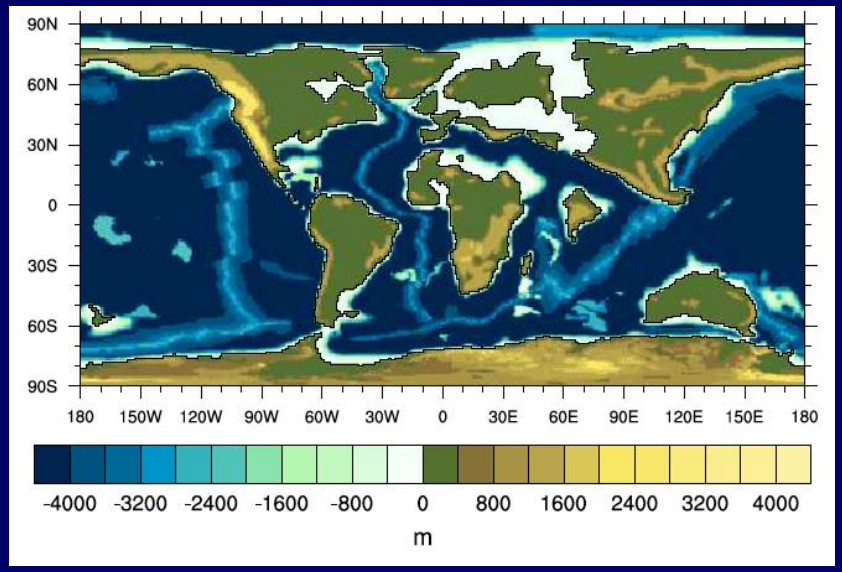
Impact of plate configuration on climate (66 – 50 Ma)

Eocene sea floor age

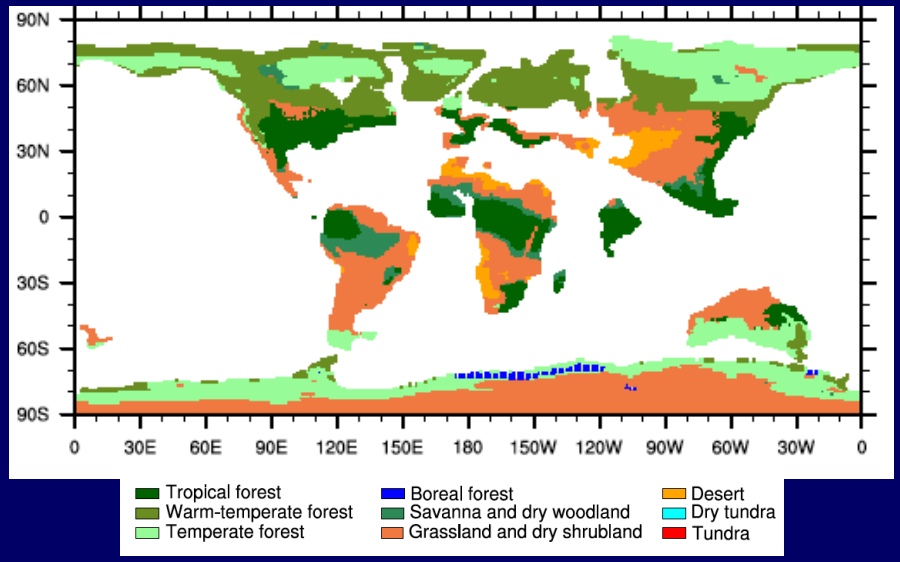


1000m ocean circulation



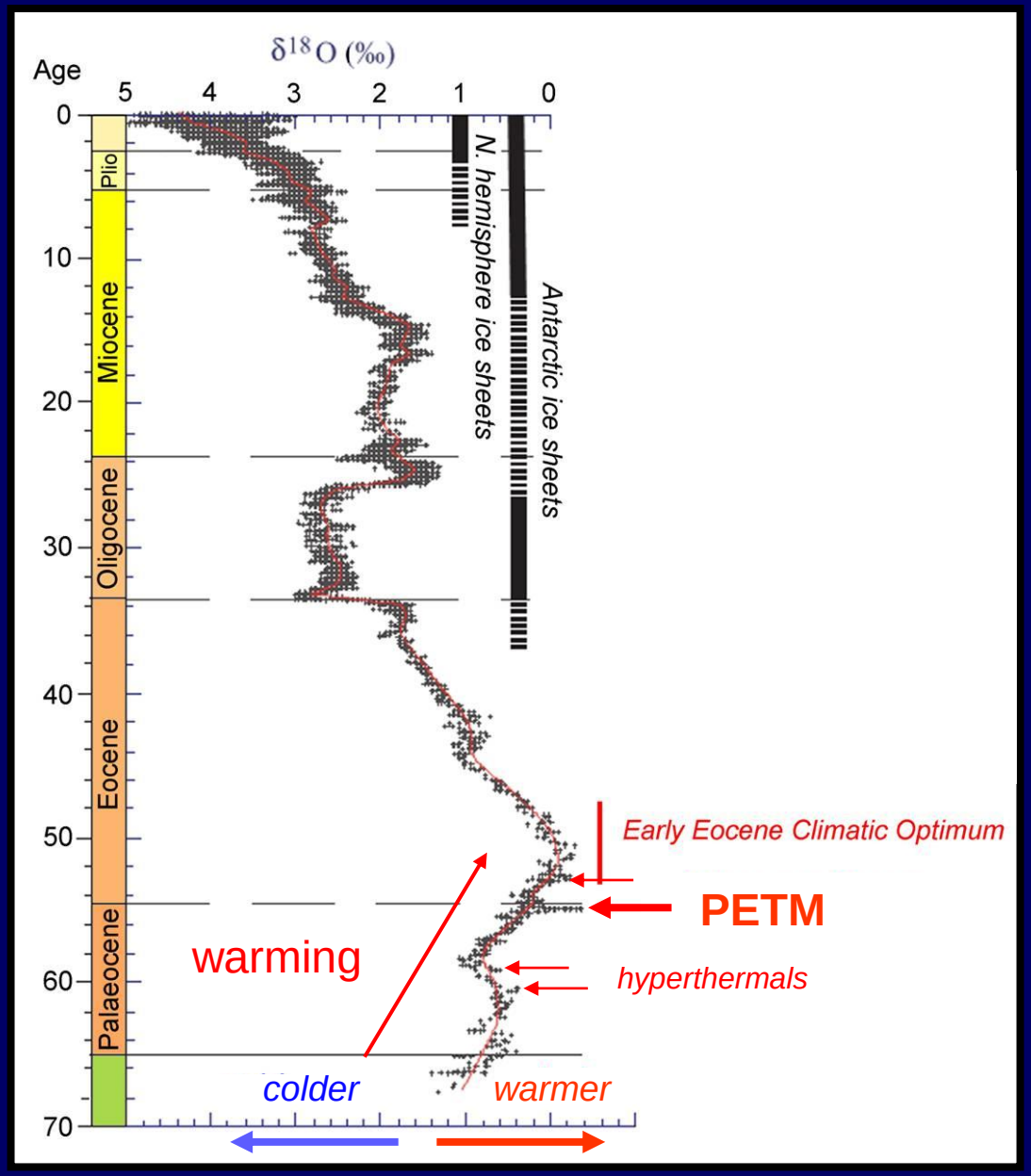


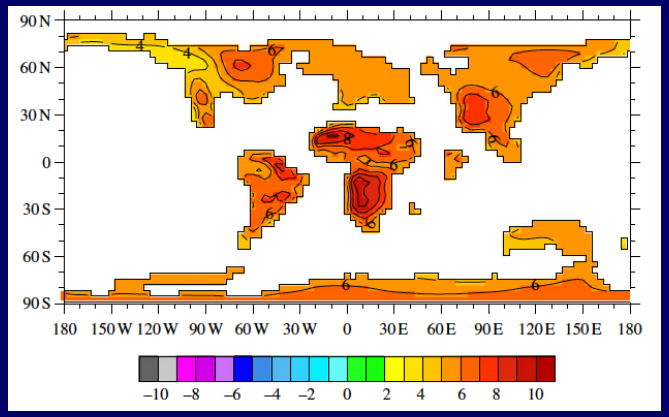
Eocene topography and bathymetry



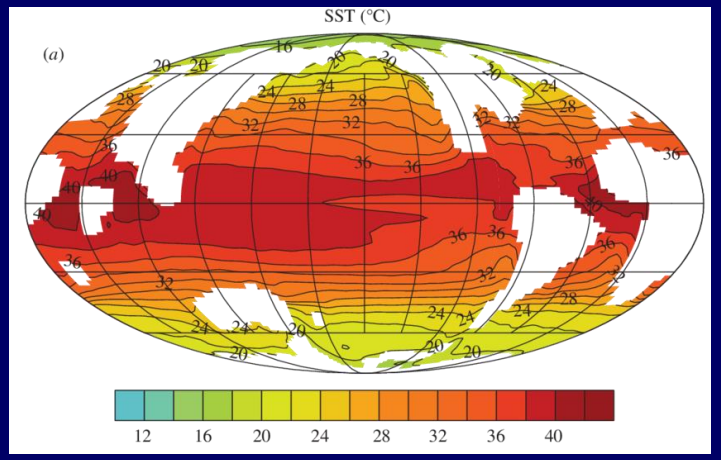
Eocene vegetation map

Paleocene-Eocene thermal maximum (PETM) – 55.5 Ma





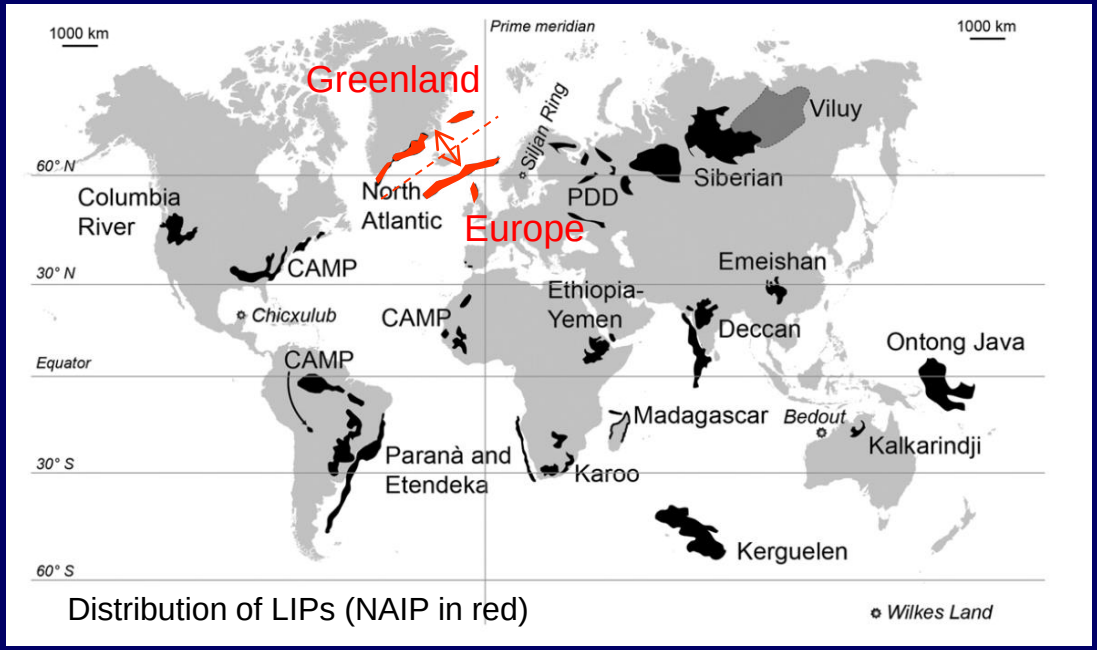
Change in air surface temperature from pre-PETM to PETM climate



Geographical distribution sea surface temperature

Implies sudden, large increase in greenhouse gas concentrations
Volcanic CO_2 insufficient source

Triggers for PETM

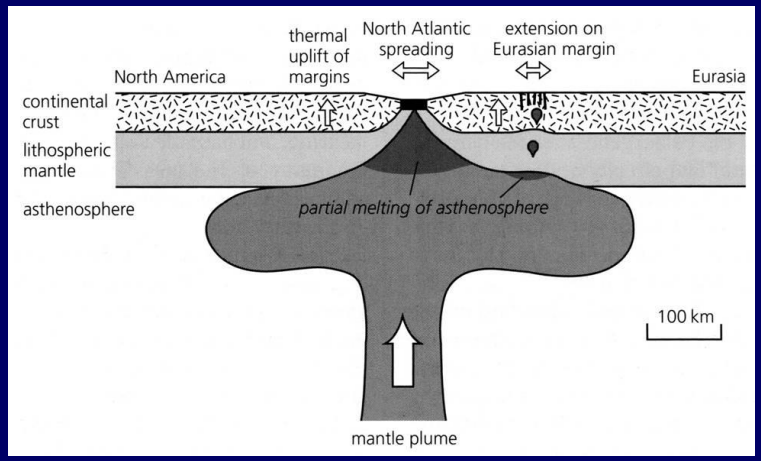


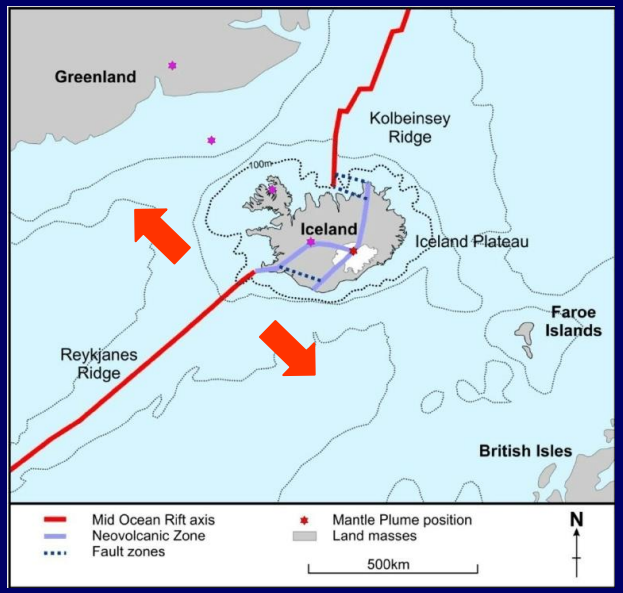
Distribution of LIPs (NAIP in red)



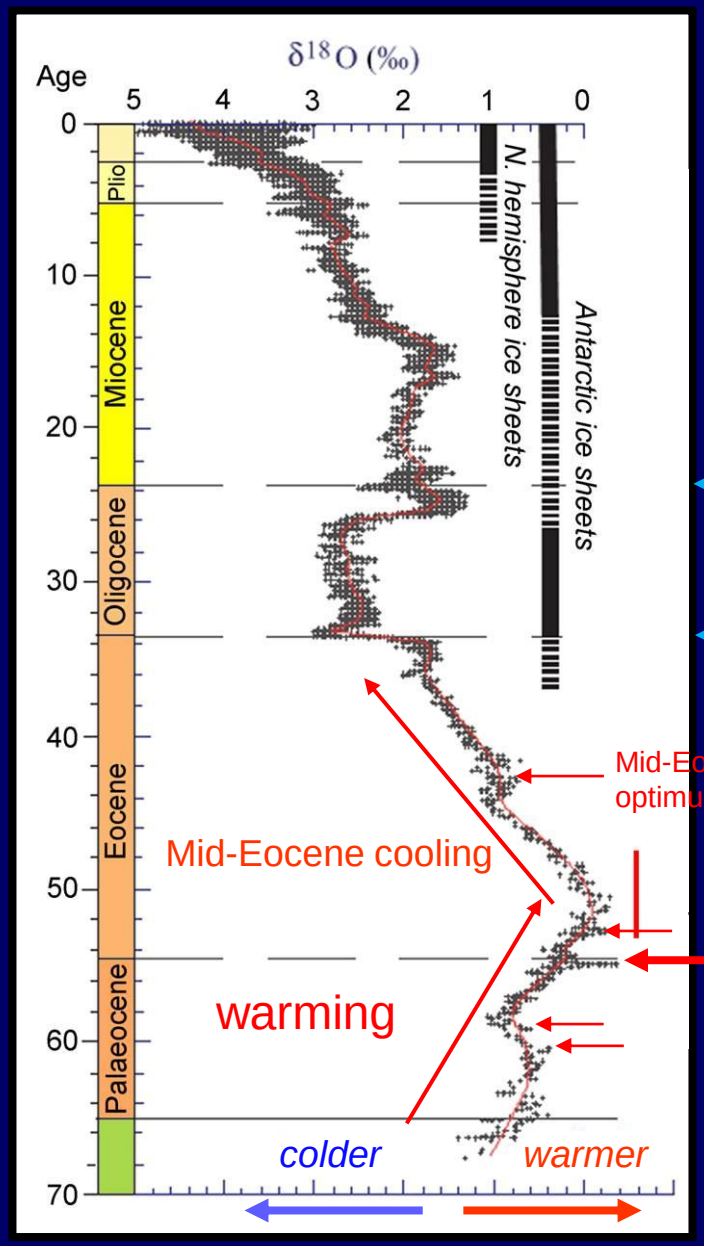
Gas hydrate-bearing sediment, from the subduction zone off Oregon

PETM is linked with main phase (56 Ma) of North Atlantic Igneous Province (NAIP)





The great Tertiary cool down



end Oligocene glaciation

end Eocene glaciation

Mid-Eocene climate optimum (MECO)

Early Eocene Climatic Optimum (EECO)

PETM

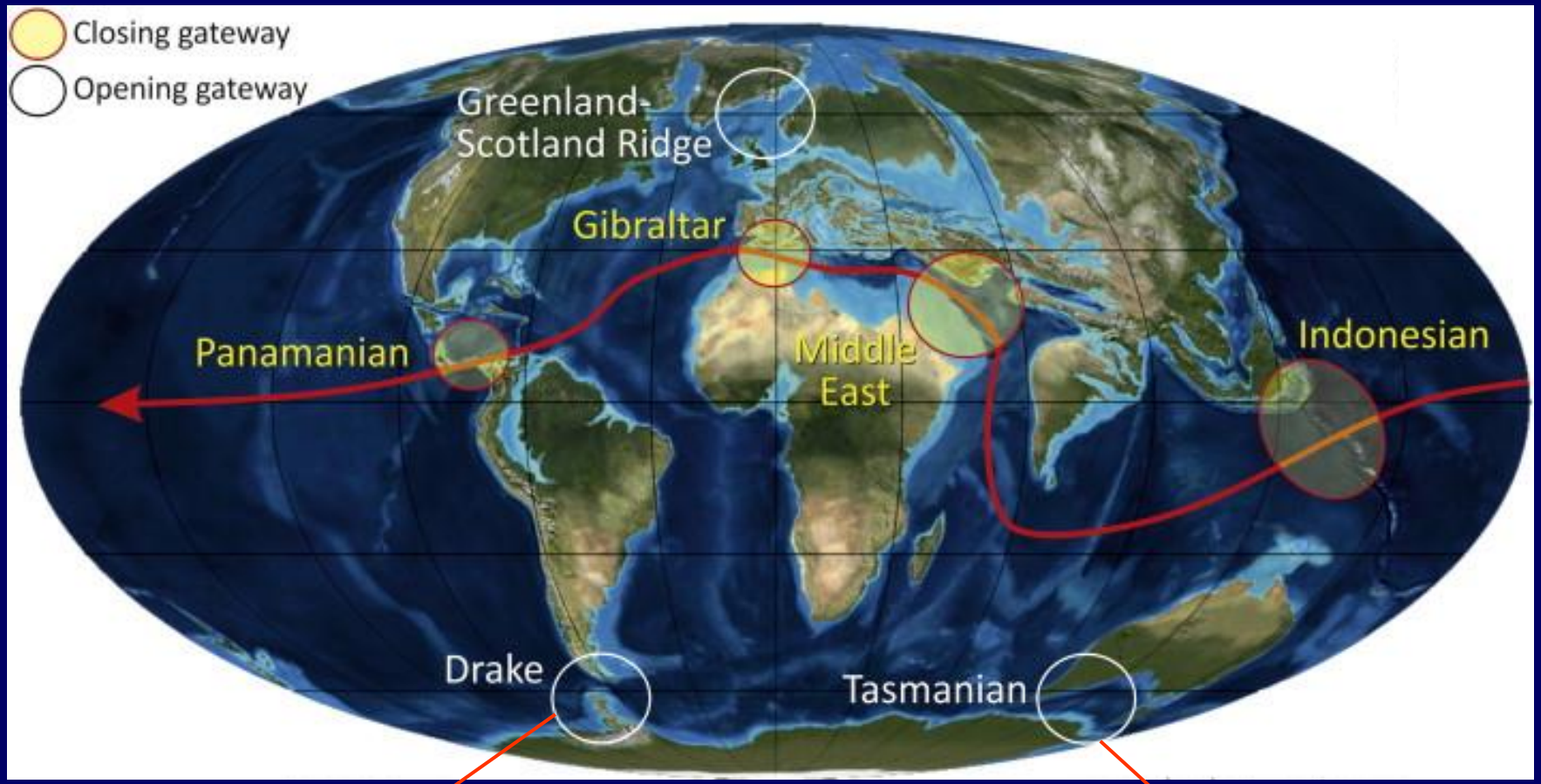
Plate reorganisation

→ circum-Antarctic ocean current

→ Andean / Tibetan uplift

Plate reorganisation → onset of Antarctic ice sheets (~35 Ma)

Eocene / Oligocene plate configuration & cooling



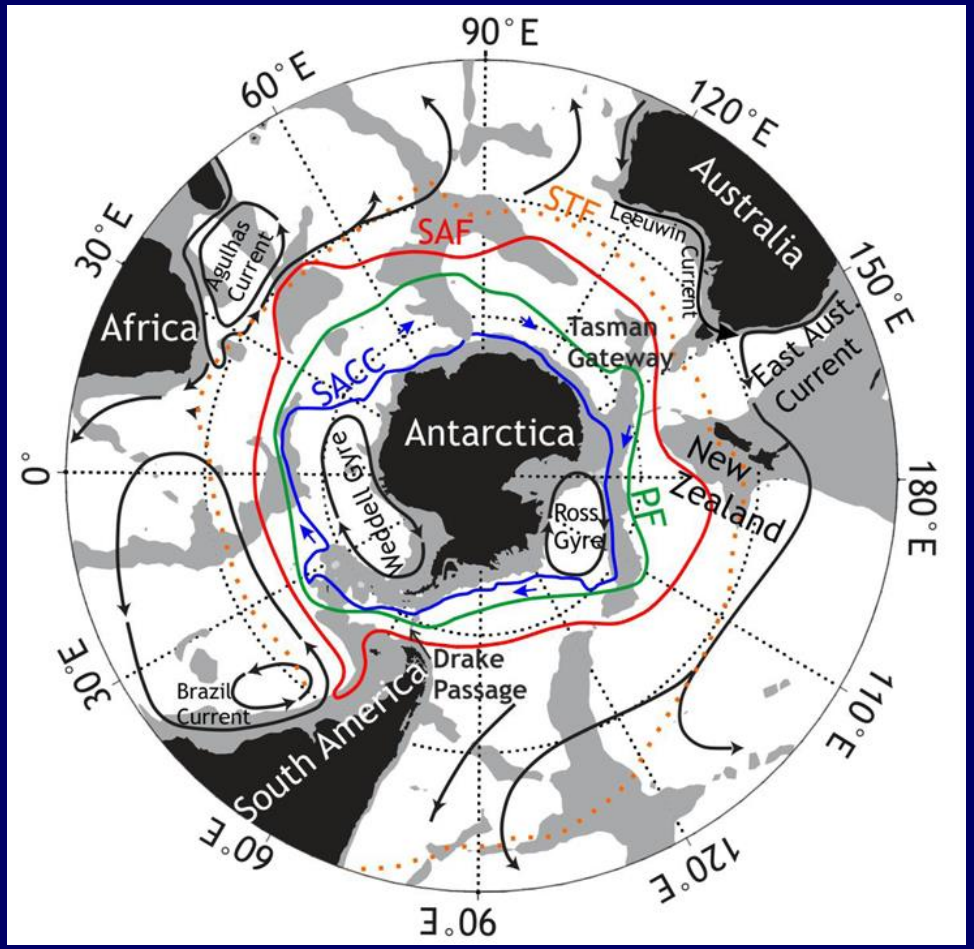
South America separated from Antarctica → Drake Passage

Australia separated from Antarctica → Tasman Passage

circum-Antarctic ocean current by 35.5 Ma

ocean circulation changed from equatorial to circumpolar

Eocene / Oligocene plate configuration & cooling



Southern Ocean circulation and circumpolar oceanic fronts

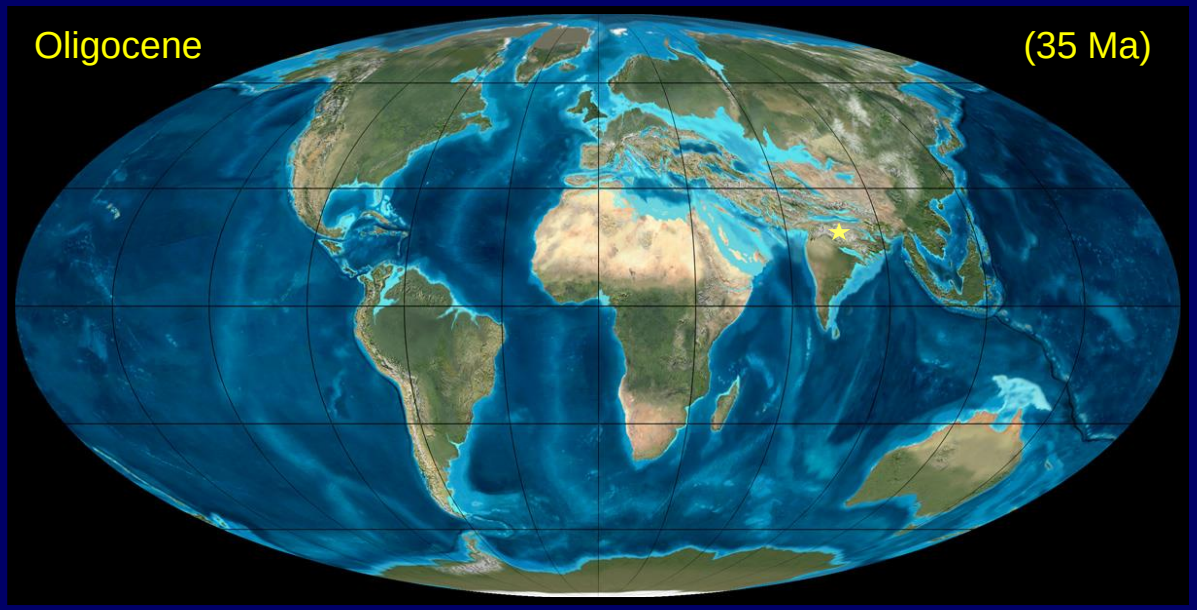
SACC = Southern Antarctic Circumpolar Current
SAF = Sub Antarctic front
PF = Polar front
STF = Sub Tropical Front

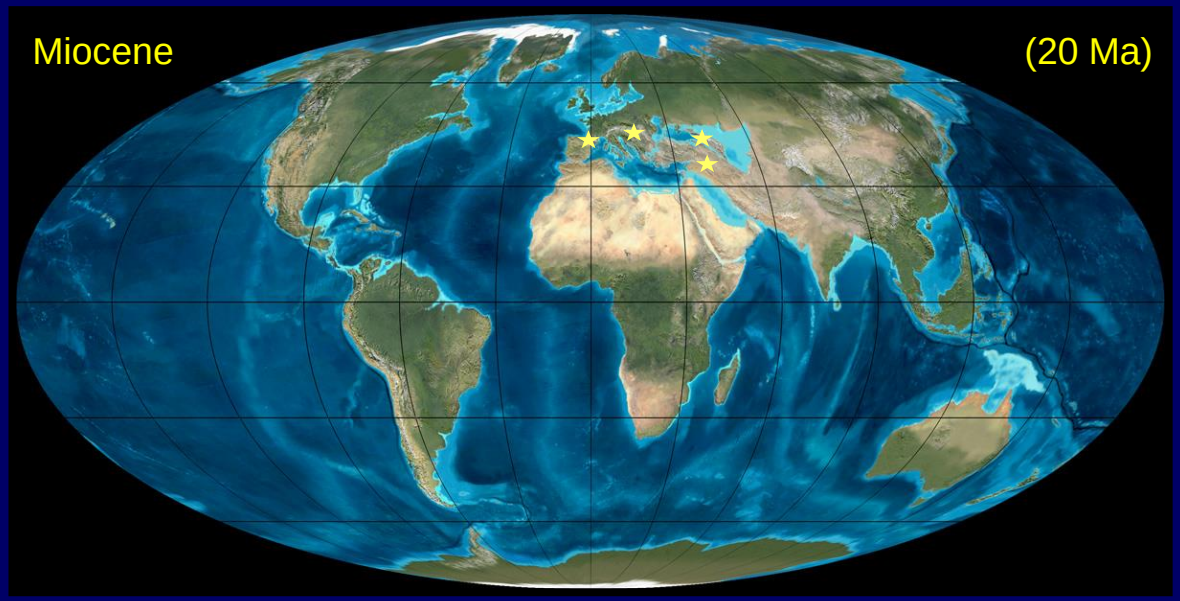
- Thermal isolation of Antarctica from warm equatorial currents
- Antarctic convergence zone – upwelling of nutrient-rich waters
- Cooling effect in middle Eocene enabled Antarctic ice sheets to form

Oligocene / Miocene plate configuration & cooling

Major mountain building

- India-Asia collision
- 55-40 Ma: uplift of Himalayas, Tibetan Plateau, Cordilleras

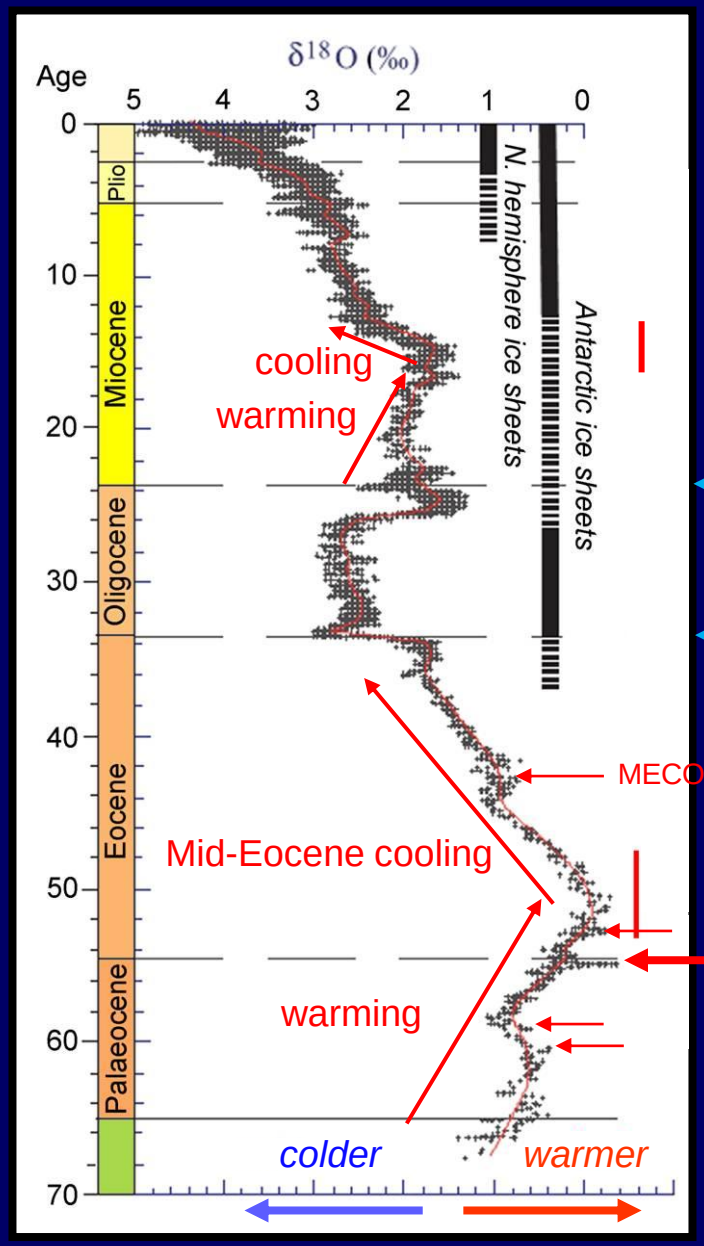




- ~35 Ma Europe / Africa convergence shifted to WNW-ESE
- 35-25 Ma: uplift of Pyrenees, Carpathians, Caucasus, Zagros

Uplift of Tibetan Plateau → enhanced weathering → cooling

The great Tertiary cool down



Intensification of AMOC

E. Antarctic ice sheet

Mid Miocene climatic optimum (MMCO)

end Oligocene glaciation

end Eocene glaciation

Panama seaway closes

Enhanced weathering and erosion from main uplift Himalayas & Tibetan Plateau

Columbia River volcanism → warming

Increased CO_2 volcanic activity

Plate reorganisation

Onset circum-Antarctic current

carbon sequestration

plate reorganisation & reduction in rate of sea floor spreading

→ cooling trend

Onset Himalayan orogeny

Short-lived (~ 200 kyr) global warming events (*e.g.*, PETM at ~ 55.5 Ma) initiated very rapidly (< 10 kyr)

N. Atlantic rifting & volcanism → warming trend

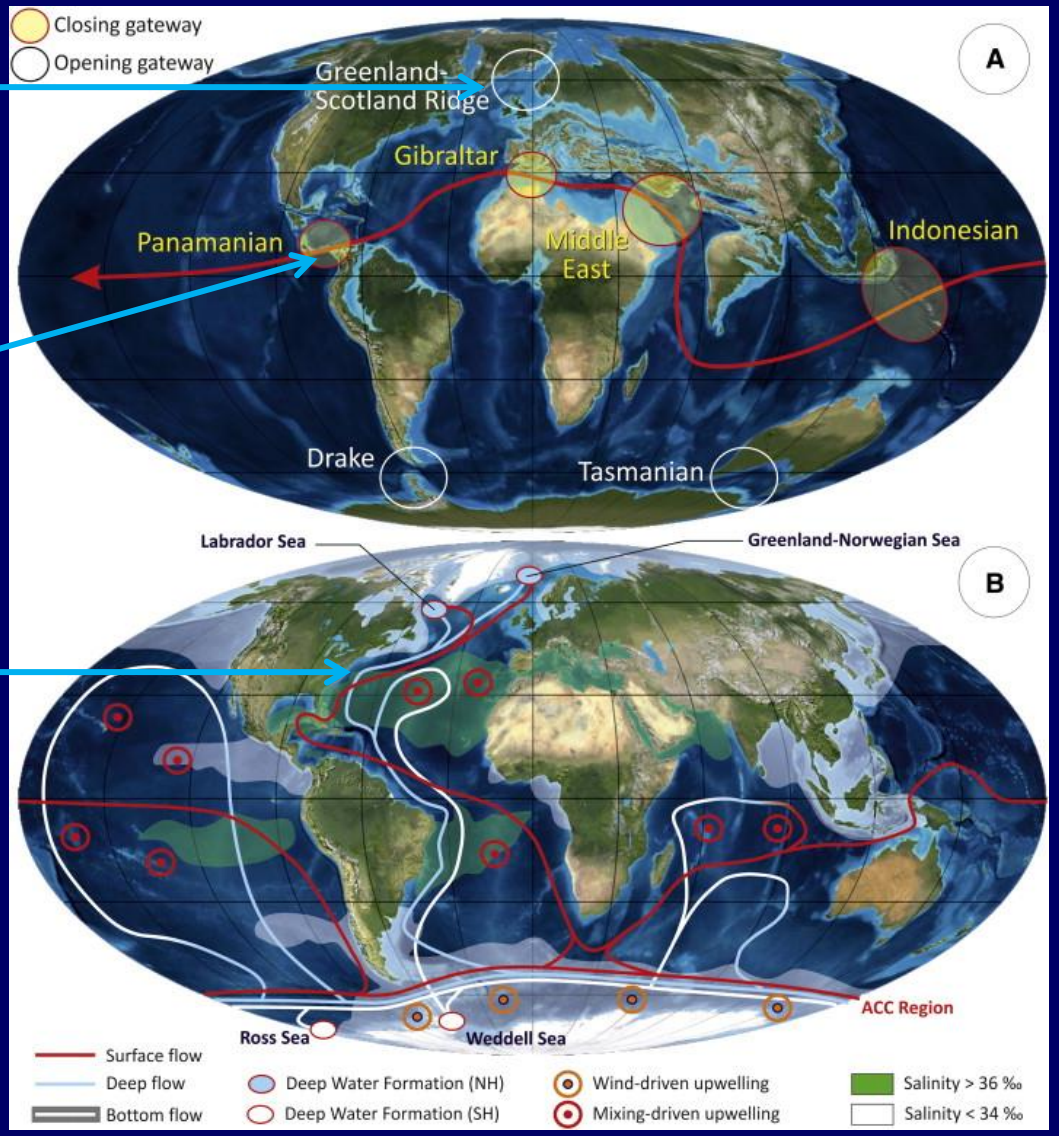
Atlantic Meridional Overturning Circulation (AMOC)

~33 Ma deep ocean connection est. between Arctic and North Atlantic

~15-13 Ma Panama Isthmus *formed and gateway closed separating Atlantic and Pacific Oceans*

Gulf stream

Delivers warm surface waters to N Atlantic



Ice masses started appearing in the Northern Hemisphere ~5 Ma, although the growth of ice sheets on Arctic land masses did not begin until after ~3 Ma when AMOC system intensified.

Onset of the Quaternary ice ages

- The following were important in establishing conditions for the onset of the Quaternary glaciations:

➢ **Position of continents allowing ice build-up at poles**

Continents at poles (Antarctic case) → growth of ice sheets OR Continents form a ring around the poles (Arctic case) → prevent sea ice dispersal (Cenozoic cooling trend)

➢ **Polar regions allowed to cool**

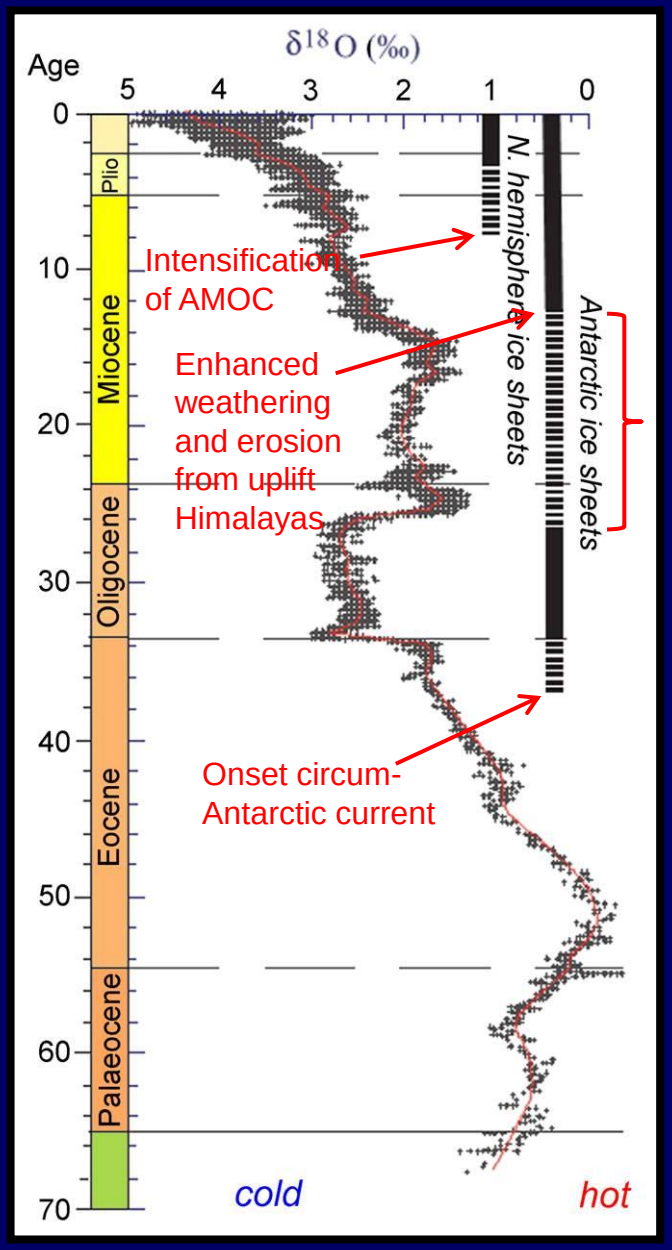
Circum-polar oceanic currents → extract heat from polar continent (opening of Tasman and Drake Passages, 35 Ma)

➢ **Relatively low levels of greenhouse gases**

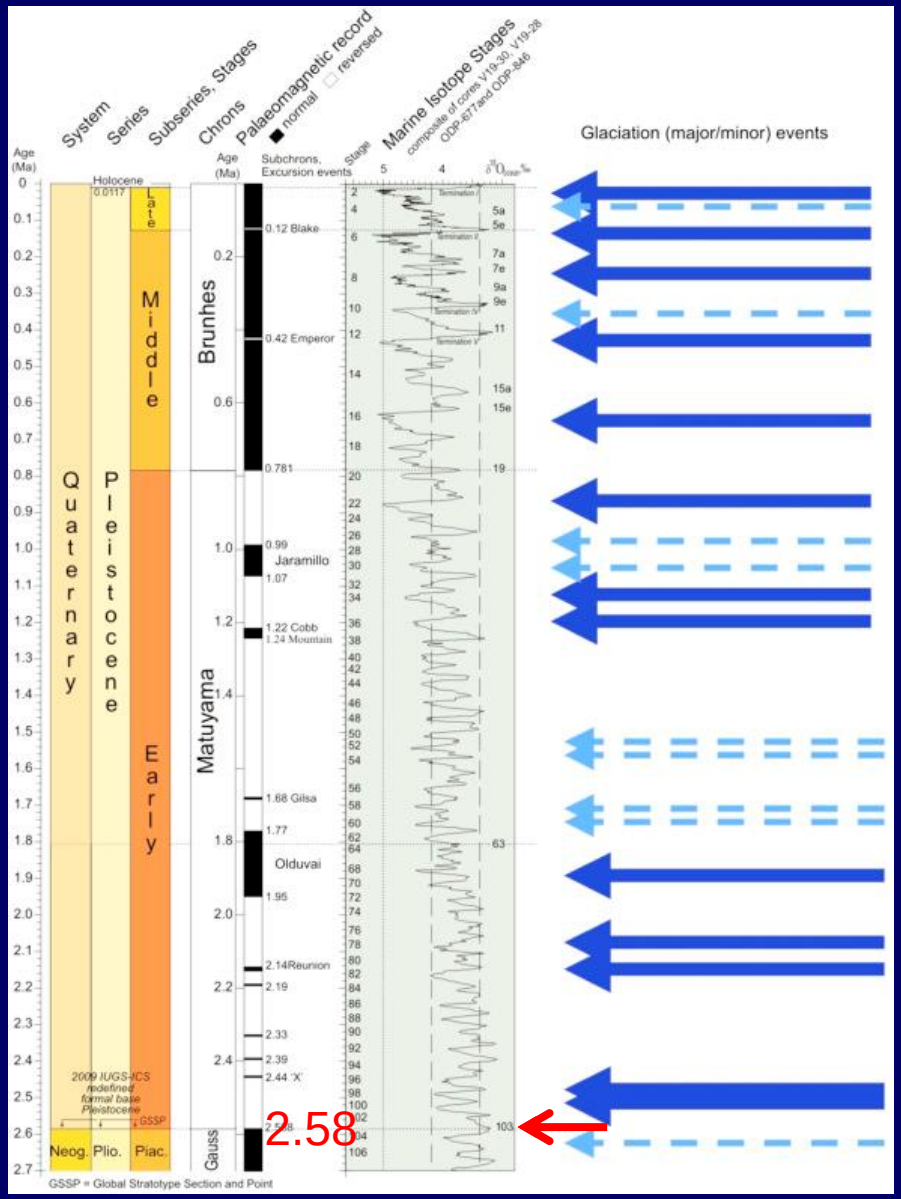
Tectonic uplift and enhanced weathering → removal of CO_2 from atmosphere (uplift of Himalayas, Miocene)

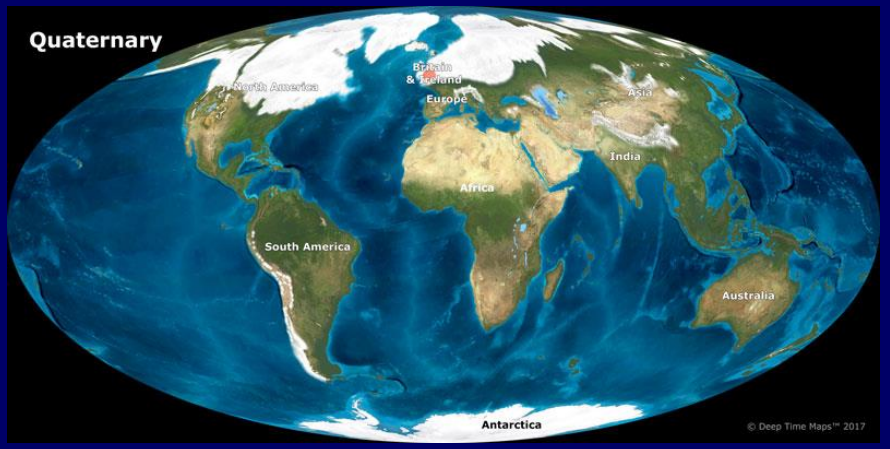
➢ **Delivery of atmospheric moisture to polar regions**

flow of shallow, warm ocean currents from the tropics to high latitudes → increased precipitation at high latitudes (intensification of the Atlantic Meridional Overturning Circulation system after closure of the Panama Gateway, 15 - 13 Ma)



About the Quaternary (2.58 Ma- 0)



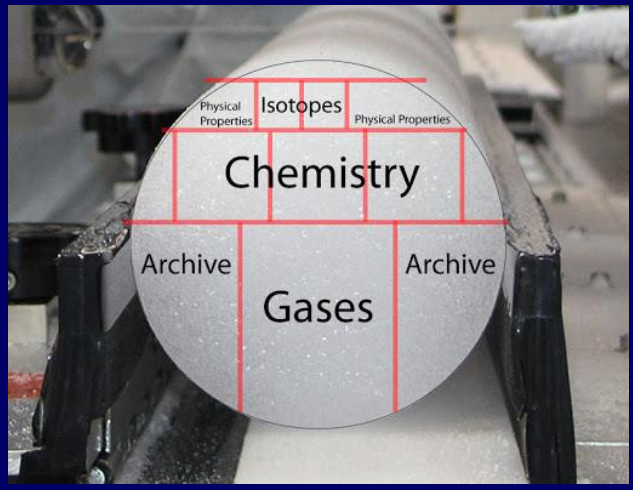


Onset of cooling: Major shift in position of North Atlantic Ocean current → large scale climate reorganisation (First major influx of ice rafted debris into mid-latitudes of Atlantic)

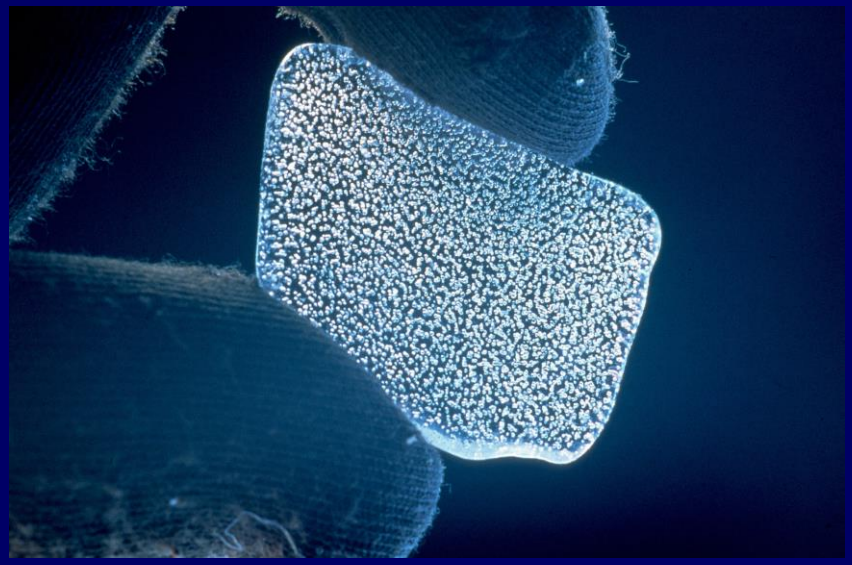
Oxygen isotopes – the ice core record



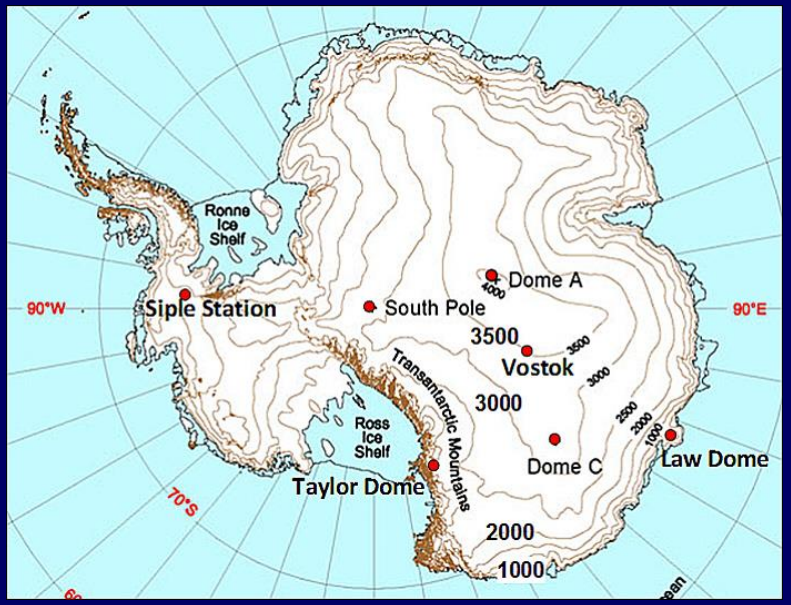
Annual alternations of darker (summer) and lighter (winter) ice



Ice core cut plan

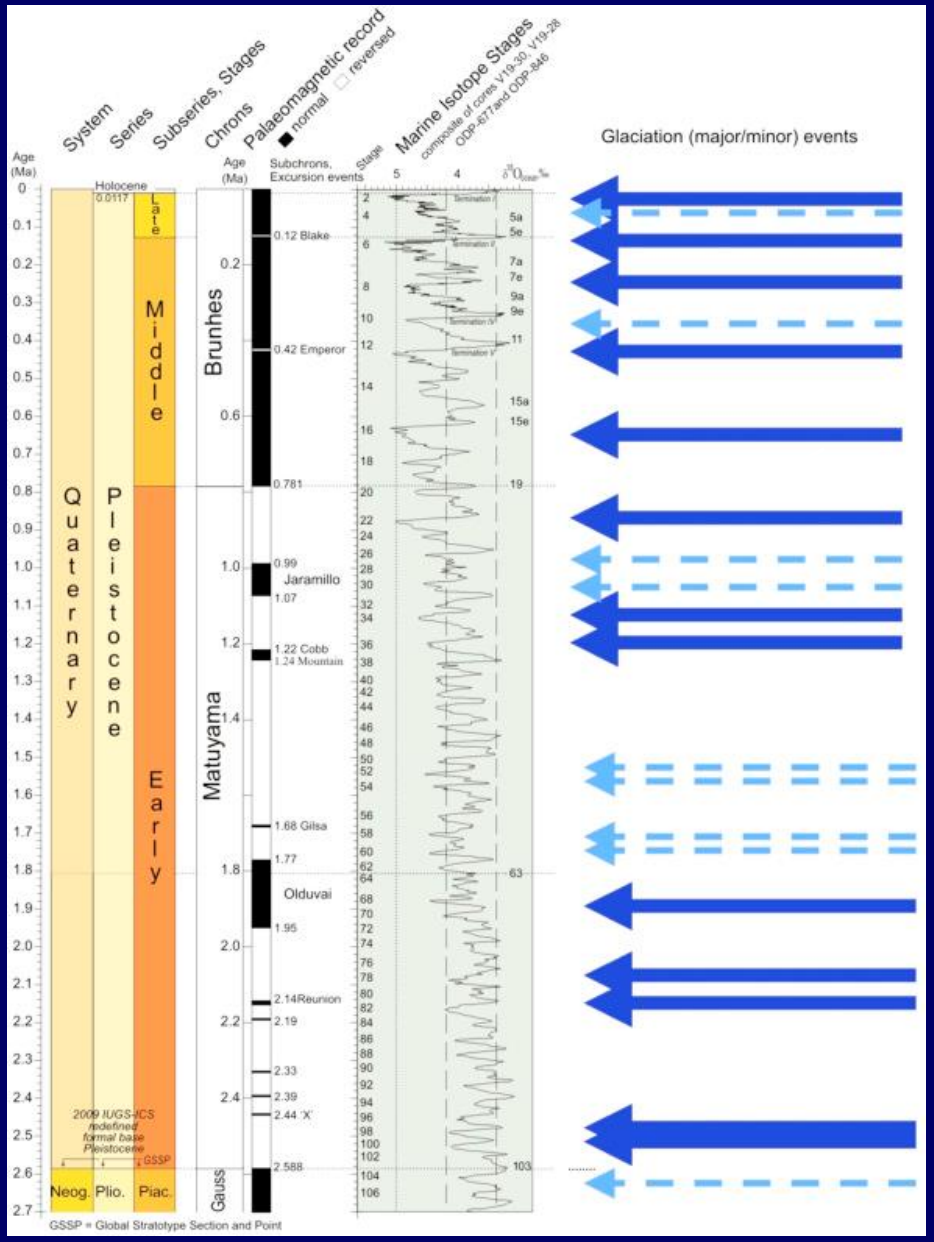


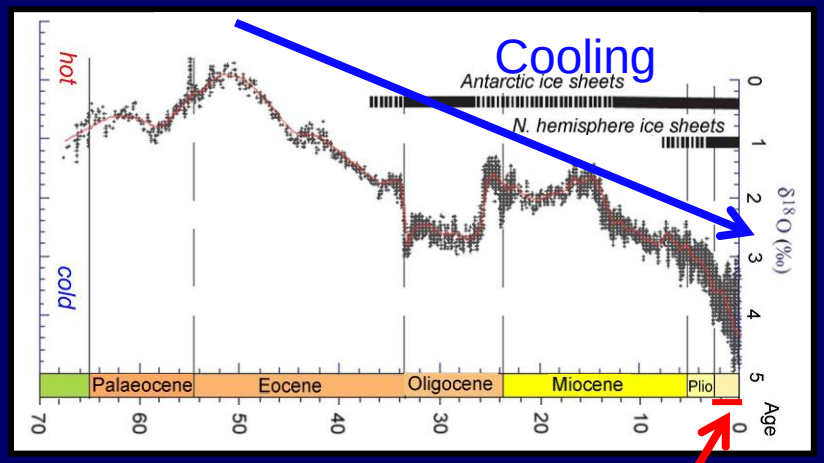
Sliver of Antarctic ice showing trapped bubbles



Ice core drilling sites on Antarctica

Pleistocene climatic oscillations

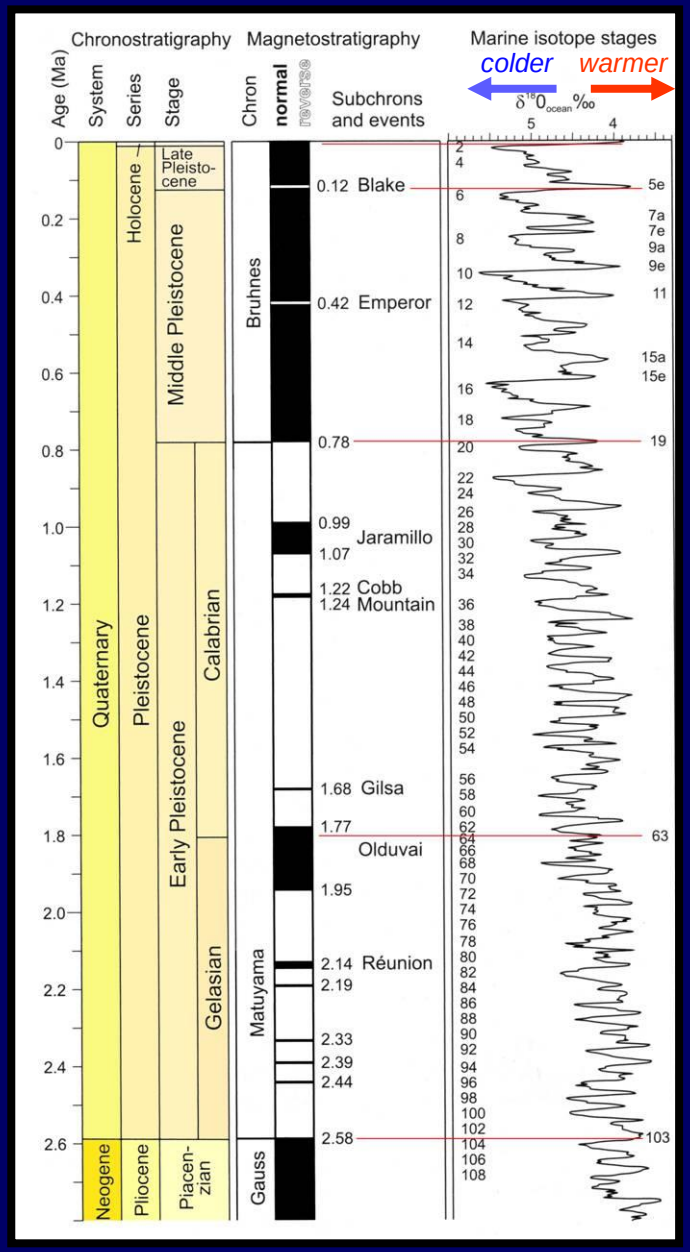




Pleistocene ice age

The oxygen isotope curve indicates ~ 50 glacial cycles through the Quaternary

Pleistocene climatic oscillations



~100 kyr cycle:

- amplitude of cycles greater
- warming much faster than cooling
- ice persisted on continents 90% of cycle

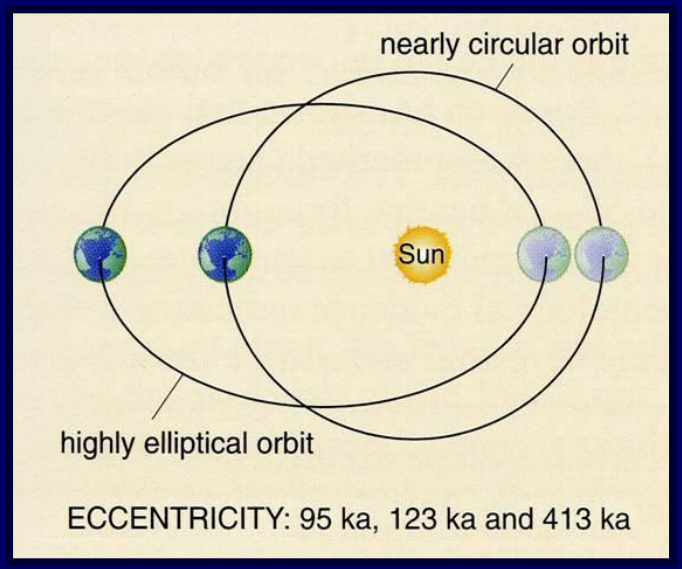
Early – Middle Pleistocene transition (EMPT)

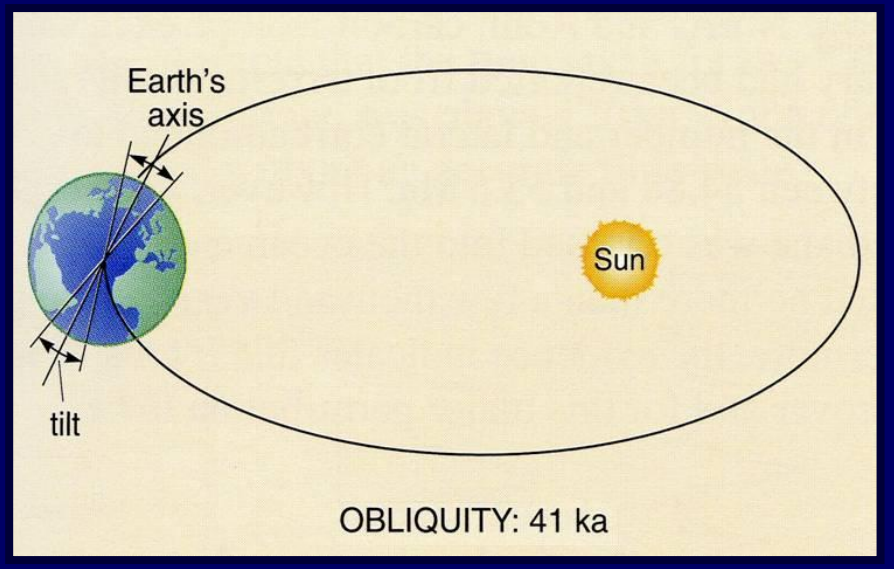
~41 kyr cycle (with a subsidiary ~20 kyr cycle):

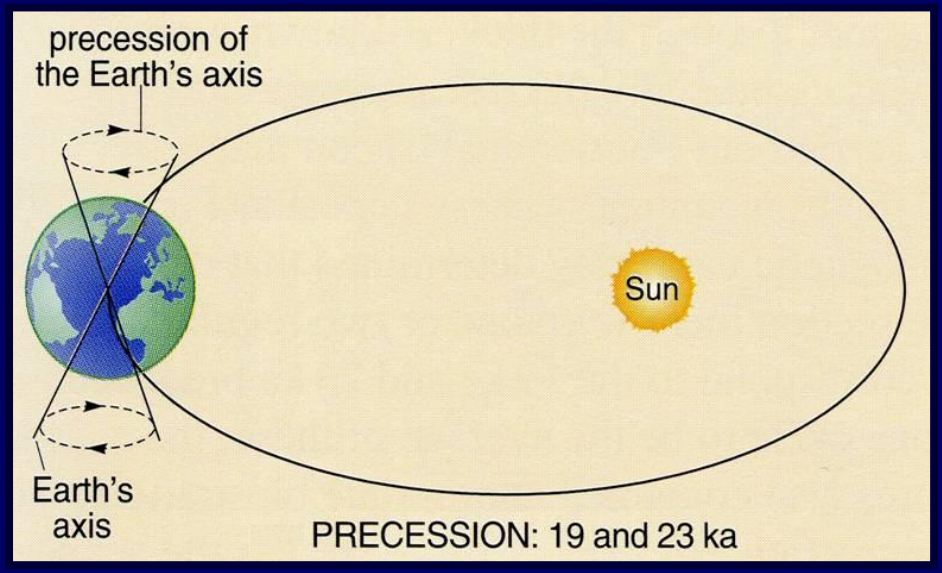
- warming phases were of similar duration to cooling phases
- ice persisted on continents ~50% of cycle

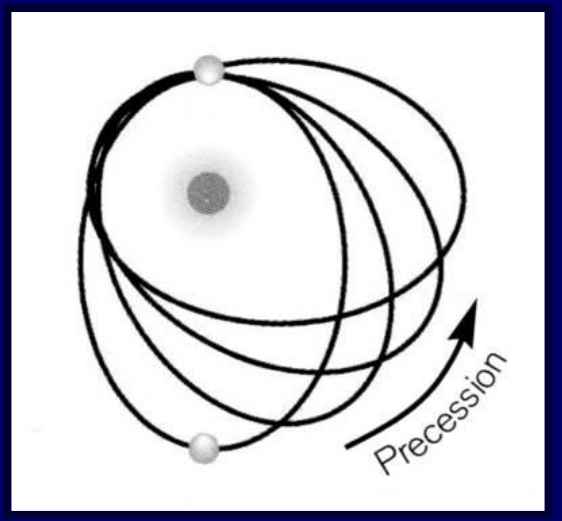
What caused these oscillations and the change in periodicity?

The pacing of the ice ages – Milanković cycles



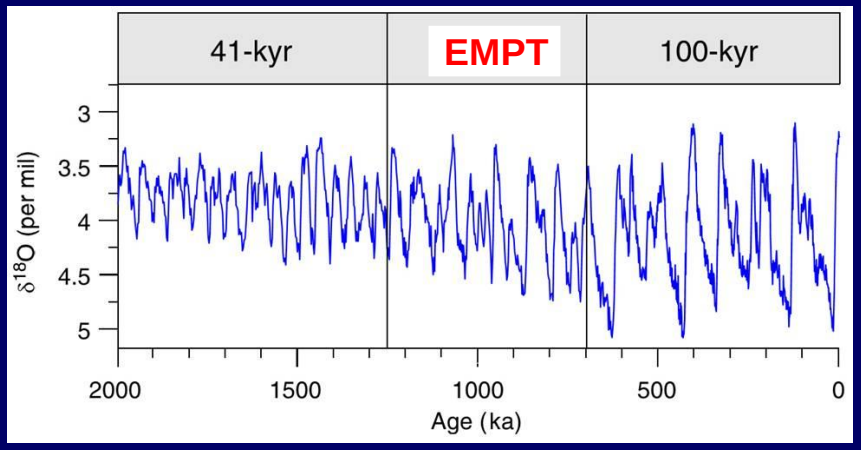


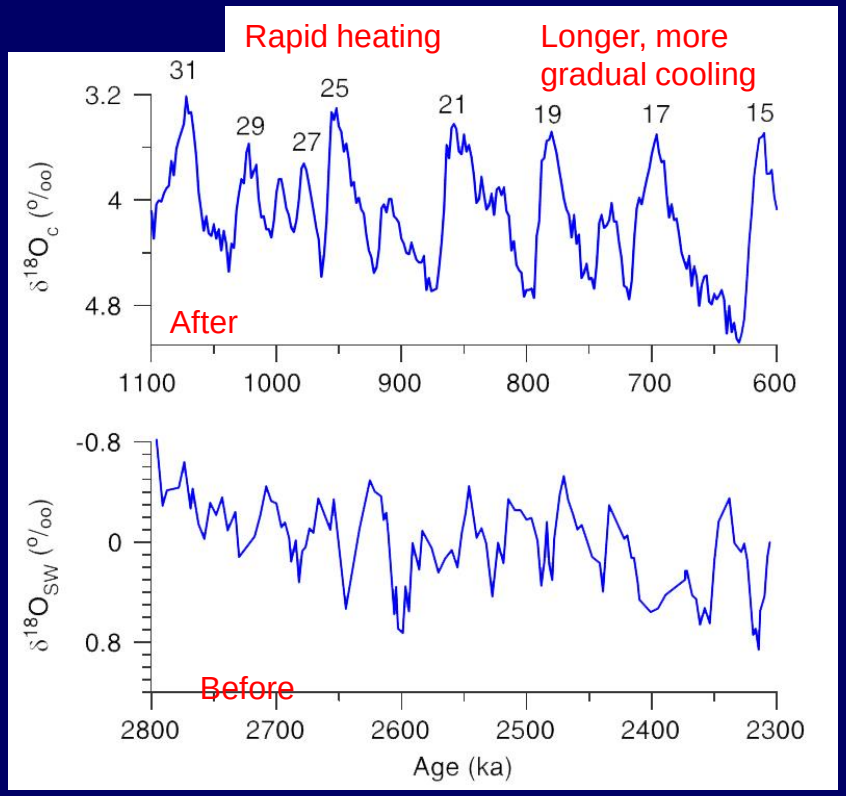




Milanković cycles

Early – Middle Pleistocene Transition (EMPT)

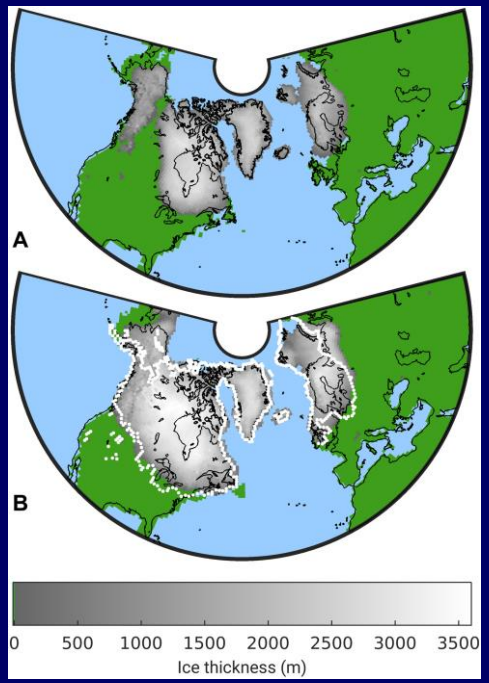




Modelled ice thickness pre-EMPT

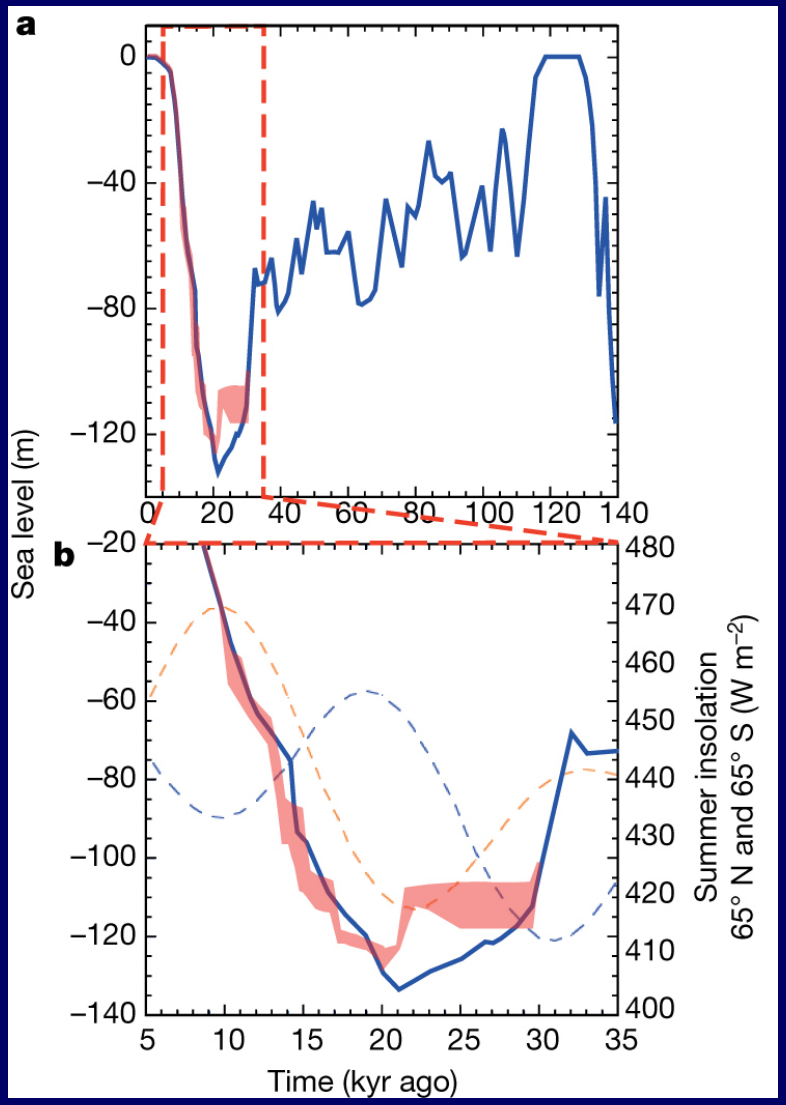
Modelled ice thickness post-EMPT

(dotted lines in B show ice extent at LGM)

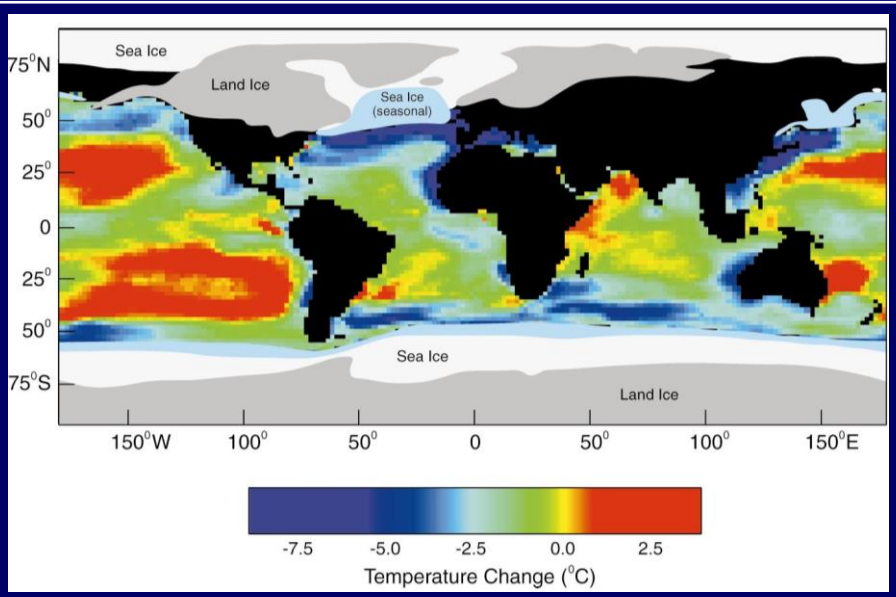


- Early Middle Pleistocene Transition (EMPT) changes in symmetry and intensification of glacial–interglacial climate cycles between about 1.2-800k.
- Low-amplitude 41-ka obliquity-forced climate cycles of the earlier Pleistocene were replaced by high-amplitude 100-ka cycles.
- These later cycles are distinctly asymmetrical and indicative of slow ice build-up and subsequent rapid melting.
- Ice sheet aerial extent before and after the EMPT was similar but ice volume afterwards was greater, and therefore the sheets were much thicker.

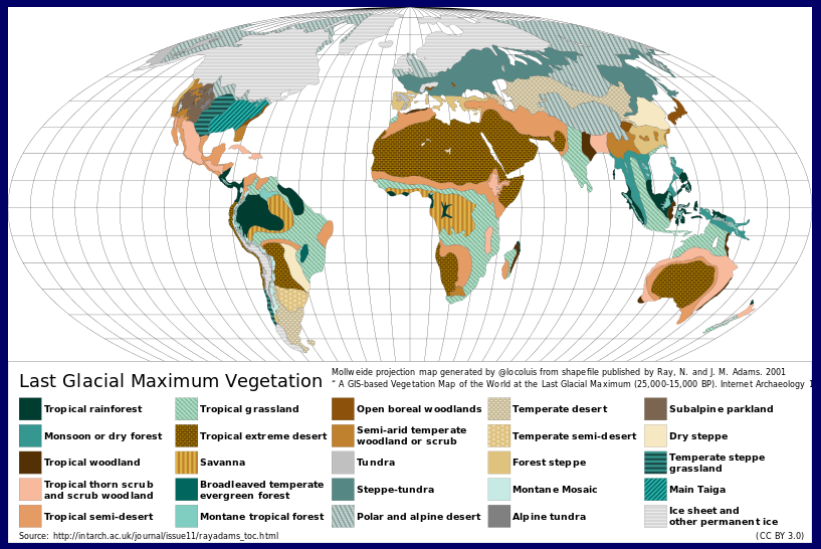
Last Glacial Maximum (LGM)



Sea level fall during the LGM

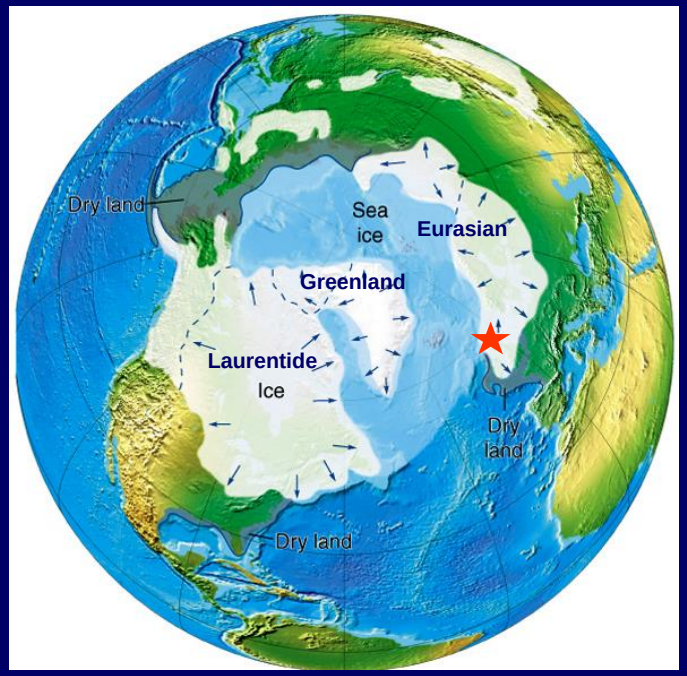


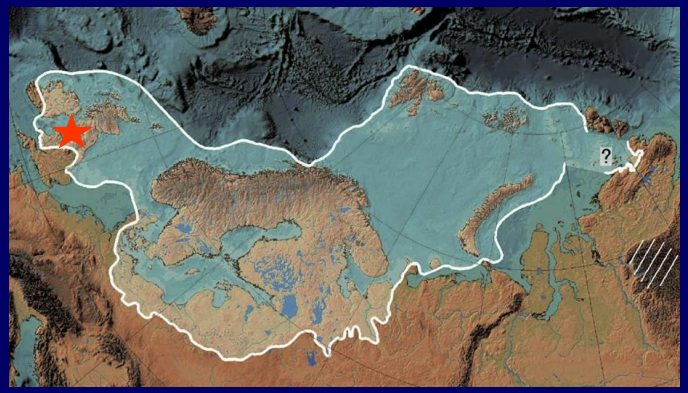
Difference in SST: LGM - preindustrial modern (CLIMAP project)



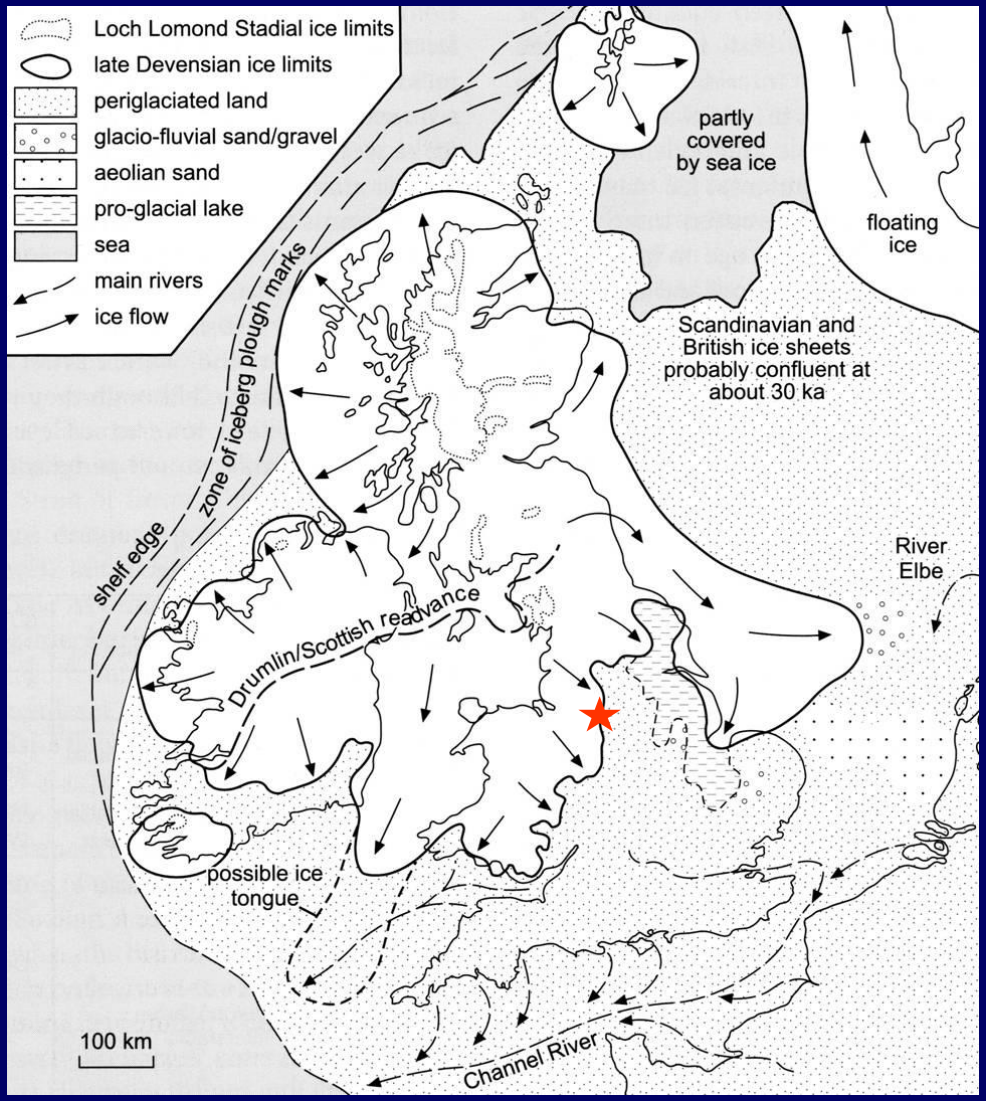
Vegetation patterns during the LGM

Last Glacial Maximum in the UK





Extent of Eurasian ice sheet at LGM (red star: location of Manchester)



The UK landscape shows abundant evidence of glacial bedrock erosion and deposition

Glacial erosion



Helvellyn, Striding Edge, and Red Tarn in the Lake District

Glacial erosion



U-shaped valley

Glacial erosion



Glacial striations on a slab shows direction of ice flow

Glacial deposits

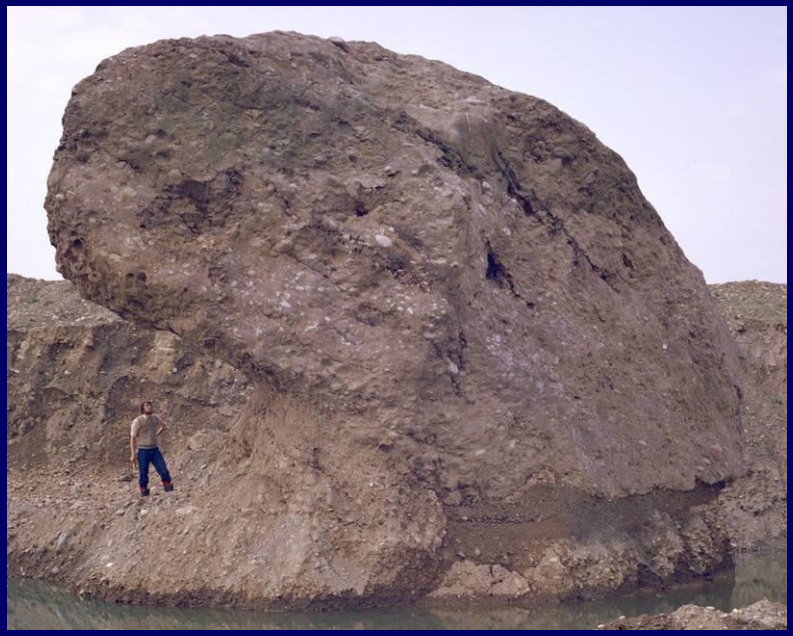




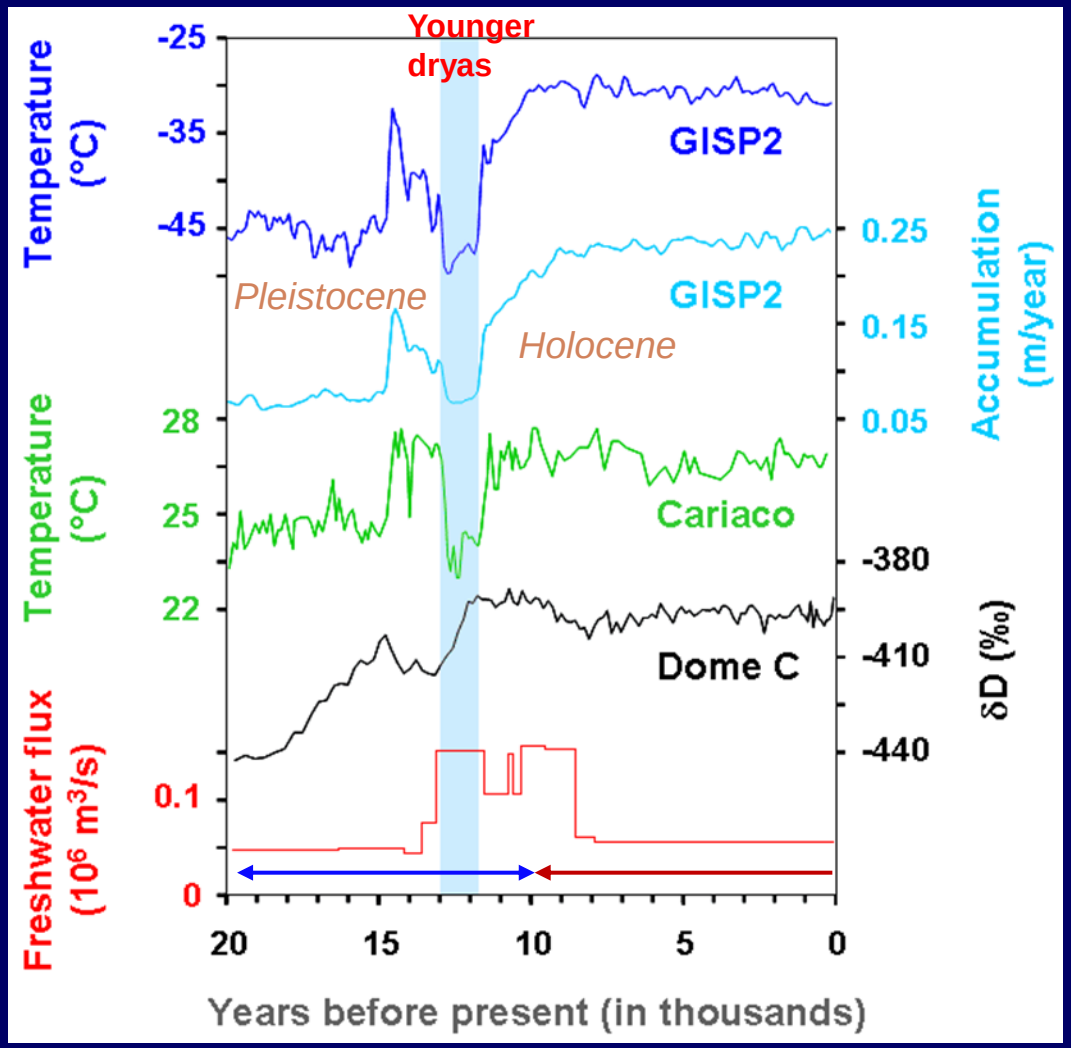
Erratics

Moraines





Younger Dryas (12.9–11.6 ka BP) climatic event



1300 yr interval of atmospheric and ocean cooling in the N hemisphere.

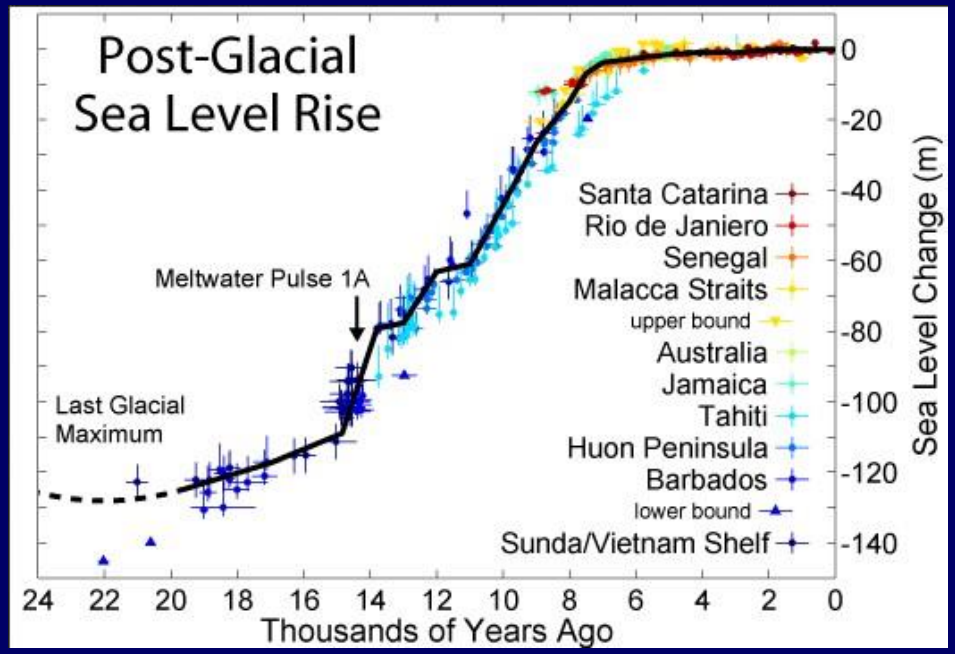
Several degrees cooling and less snow accumulation based on proxy records.

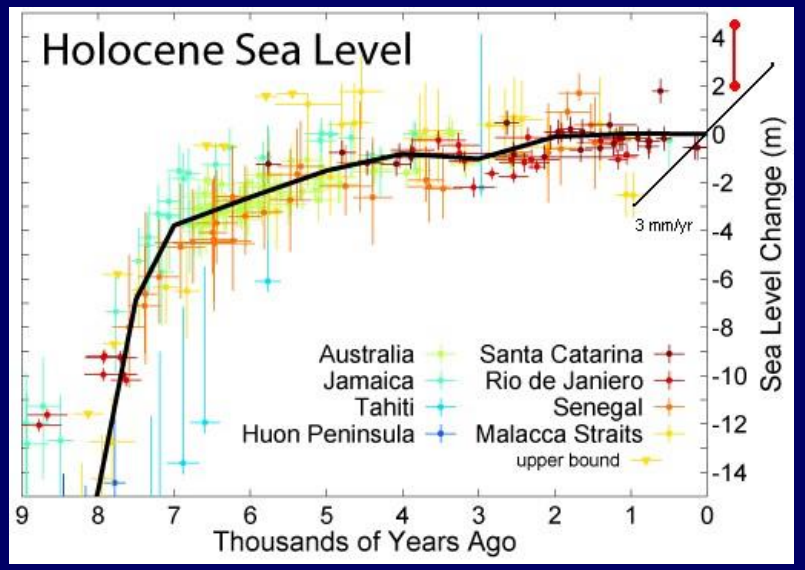
S hemisphere continued to warm.

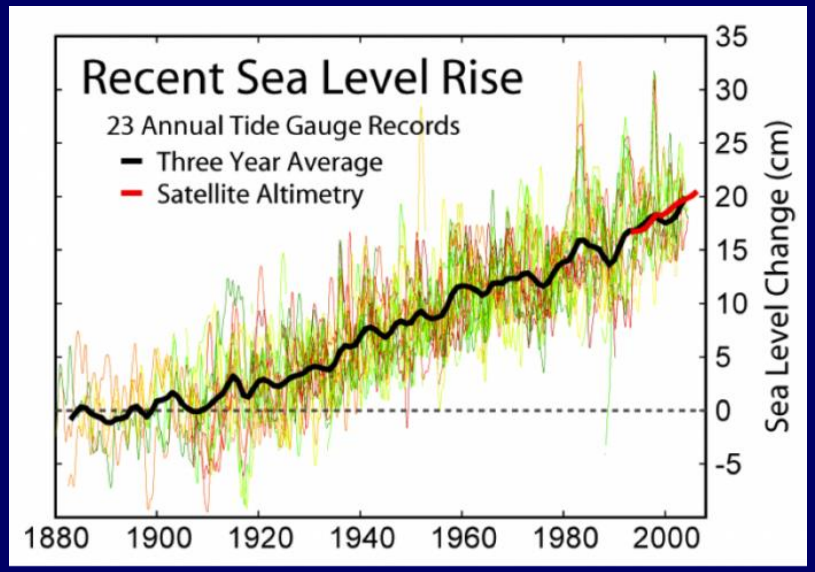
Main hypothesis:
Catastrophic drainage of Lake Agassiz (glacial lake in US), leading to freshwater-induced Atlantic Meridional Overturning Circulation (AMOC) collapse and abrupt reduction of the associated northward heat transport.

Alternative hypotheses:
Extraterrestrial impact or volcanic – leading to increase atmospheric dust and cooling.

Holocene sea level rise







Eustatic sea level rise (rise in global sea level due to melting of the polar ice sheets) since the LGM.

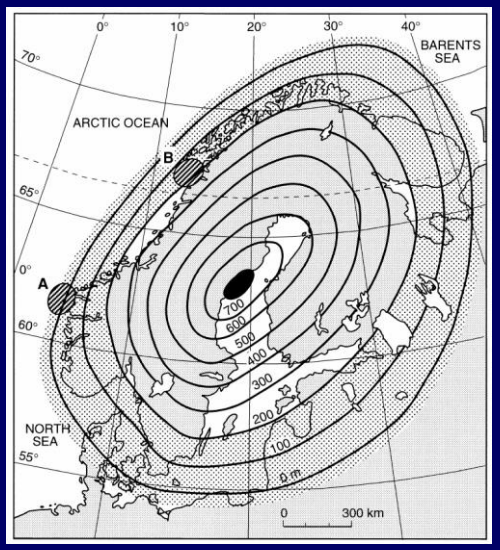
Net effect is complex interplay between:

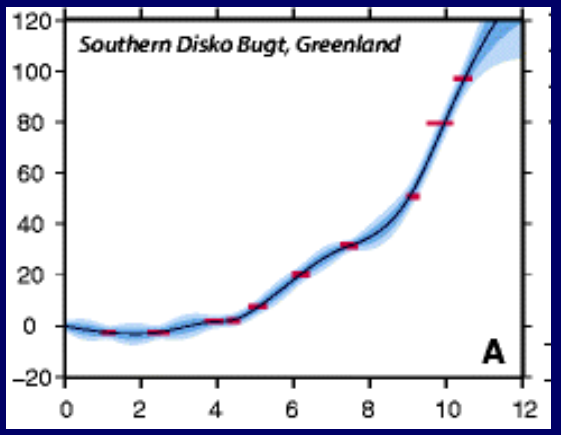
- eustatic SL rise (+130 m)
- isostatic rebound (glacio- and hydro-)
- ocean dynamics, tectonics & local factors e.g. Tidal regime change)

Variability in Holocene sea level rise: near field vs far field



Near field (glaciated areas at LGM)
e.g., Scotland, Greenland, Canada

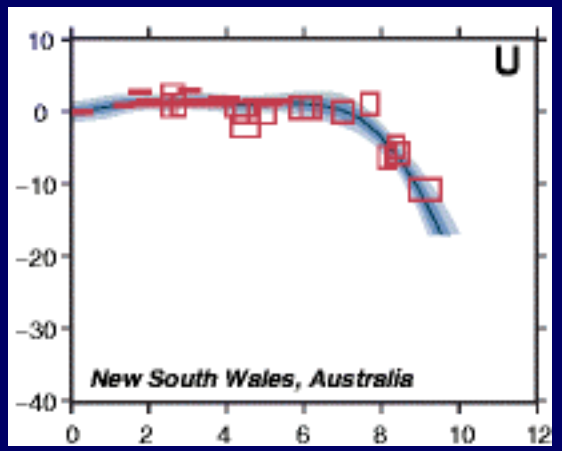




Affected by post glacial rebound (rheology)



Far field (areas far from ice sheets)

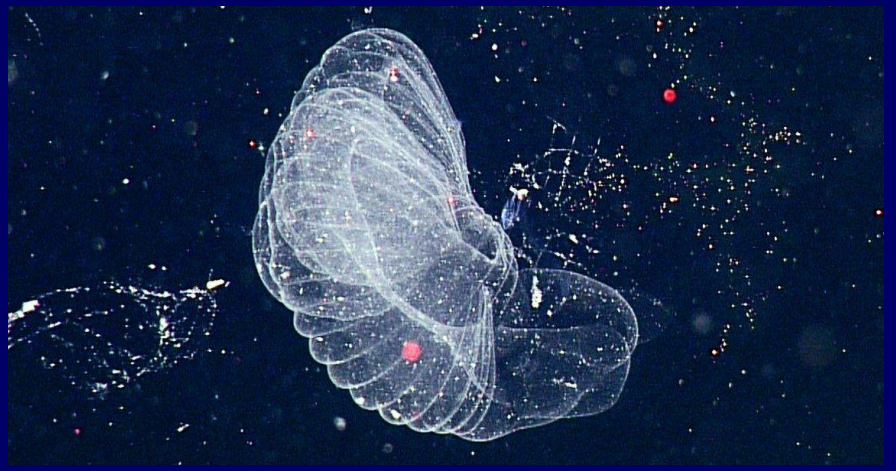


Constrain eustatic sea level change

Anthropocene: distinctive signals







Microplastics have moved into every crevice on Earth



Plastiglomerates from Hawaii

When did it start?

- Mid-Holocene when agricultural practices started to influence atmospheric composition
- AD 1750 when the Industrial Revolution gathered momentum
- AD 1950 when population growth, industrialization, and the use of atomic fission accelerated

How long will it last? Is it just an event horizon?

Summary of main events

- The Cenozoic era (0-66 Ma) has a well-preserved geological record
- Climate switched from greenhouse conditions to icehouse conditions at the present day as recorded by oxygen isotopes in benthic microfossils.
- Max. temperatures occurred 50 Ma ago during the Eocene Climatic Optimum caused by Pacific Ocean circulation carrying warm waters to high latitudes.
- PETM hyperthemal event (55.5 Ma) related to volcanic CO_2 from North Atlantic Igneous Province, ocean acidification and methane release (gas hydrates)
- First continental icesheets present at end-Eocene (35 Ma ago) due to thermal isolation of Antarctica by ACC from opening of oceanic gateways.
- Permanent southern polar ice sheets at 32 Ma resulting from atmospheric CO_2 drawdown by Himalayan uplift
- Arctic ice sheets established by 3 Ma ago by intensification of the AMOC (closure of oceanic gateways) leading to increases precipitation in polar regions.
- Quaternary marine and ice core oxygen isotope records of glaciations related to Earth's orbital variations in 100 Ka (eccentricity) and 41 Ka (obliquity) cycles with transition (EMPT).
- Since the Last Glacial Maximum (LGM) a combination of eustatic sea level rise (global = 130 m) and isostatic rebound (local = glacial unloading of land surface).

Exam Guidance: How to Build a Planet

- Two hour exam in January
- A combination of multiple choice, multiple answer, true/false, matching and ordering.
- The question order is randomised.
- You will not need a calculator to answer the questions in this test.
- The examination consists of 100 questions. Answer them all.
- Marks will not be deducted for incorrect answers.
- Each question is worth 1 point, unless indicated otherwise on a question.
- Read the questions carefully and follow the instructions given in the question.
- Each question has only one correct answer unless a question instructs you to order, match or select more than one answer. If a question asks you to select two correct answers but you select three, you will be awarded no marks for that question.
- There is no partial credit for questions with more than one answer